Factor Investing in Emerging Markets: From Multi-Factor Industry Portfolios to a Market-Neutral Alpha Strategy

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Abstract

This paper studies a systematic six-factor strategy for MSCI Emerging Market equities using a strict temporal split between design and evaluation. Equal-weighting six signals—size, value, quality, momentum, volatility, and dividend yield—produces the highest average industry-level Sharpe ratio in the factor-selection window without parameterized signal weighting. In the portfolio-construction window, turnover-penalized mean-variance optimization (TO_MVO) attains the highest transaction-cost-adjusted Sharpe ratio while reducing total monthly turnover to 19.2%, compared with 28.0% for momentum weighting. In the matched January 2019–September 2024 out-of-sample window used for benchmarked and hedged comparisons, the long-only strategy records a gross Sharpe ratio of 0.607 versus 0.356 for EEM, while the EEM-hedged strategy records a gross Sharpe ratio of 0.828, a net Sharpe ratio of 0.673, Jensen's alpha of 4.16% per year, and a US diversification premium of 0.642. Holdings-based diagnostics over 2021–2025 suggest that the 2022–2023 weakness coincided with a more challenging stock-selection environment, and that the positive CMA loading in multivariate factor regressions should be interpreted cautiously in light of CMA–HML collinearity.

Keywords: Factor investing, emerging markets, multi-factor portfolios, turnover-penalized optimization, beta hedging, information coefficient, US diversification

JEL: G11, G12, G15

[0]*Corresponding author: Andy Zhang, Columbia Business School, 3022 Broadway, New York, NY 10027. E-mail: zz3231@gsb.columbia.edu. Replication code is available at `github.com/zz3231/em-factor-investing`. The authors thank Professor Yin Luo for invaluable guidance and helpful discussions throughout this project, and Columbia Business School for research support. All data are from Refinitiv Datastream, MSCI, iShares (EEM) via CRSP, and the Ken French Data Library. The authors declare no conflicts of interest.

1 Introduction

Factor investing has transformed quantitative asset management, yet its application in emerging markets (EM) remains underdeveloped relative to the opportunity. The MSCI Emerging Markets universe represents roughly $7–8 trillion of investable equity capitalization across more than twenty countries and accounts for approximately 13% of global market capitalization, yet attracts a disproportionately small share of quantitative research. Lower analyst coverage, weaker information efficiency, wider dispersion of accounting practices, and greater institutional frictions create a setting in which systematic, signal-based strategies are especially relevant (Cakici et al., 2013; Hanauer and Linhart, 2015; Hou et al., 2011). The academic literature confirms that canonical factor premia—size, value, momentum, and quality—exist internationally, including in EM (Fama and French, 2012, 2015). The practitioner challenge is translating that evidence into an implementable strategy that survives the reality of high transaction costs, thin liquidity, and the EM researcher's perennial adversary: data snooping.

This paper reports the complete pipeline from signal to market-neutral portfolio. We examine six stock-level characteristics across eleven GICS-based industry groups within the MSCI EM universe over January 2004 to March 2025. Critically, we impose a strict three-stage temporal separation that eliminates look-ahead bias in every design decision: factor composition is selected on 2009–2013 data, portfolio construction is selected on 2014–2018 data, and out-of-sample evaluation begins in January 2019. The matched benchmark, hedging, and factor-model comparisons use the 69 months through September 2024 for which EEM total-return data are available in the working dataset, while the underlying stock-level signal panel and holdings diagnostics extend through March 2025. This structure directly addresses the multiple-testing critique of Harvey et al. (2016), the factor-decay evidence of McLean and Pontiff (2016), and the data-snooping concerns documented by Butler and Kwon (2021).

In the matched January 2019–September 2024 comparison window, a strategy built on transparent design choices delivers a hedged gross Sharpe ratio of 0.828 (0.673 net of conservative 45-basis-point transaction costs), Jensen's alpha of 4.16% per year, and a maximum drawdown of 5.2%. The strategy also exceeds the hurdle Sharpe criterion for US portfolio diversification by 0.642, aided by its 0.223 correlation with US equities.

Principal Findings

Our five principal findings are as follows.

1. **A simple equal-weighted composite compares favorably in factor selection.** Equal-weighting all six factors (All6-EW) achieves the highest average Sharpe ratio across industries (0.859) during the 2009–2013 selection period with zero degrees of freedom. More

complex composition methods—IC-weighted composites, correlation-filtered subsets, and rolling single-factor selection—consistently underperform, consistent with the estimation-error argument of DeMiguel et al. (2009).

2. **Turnover penalization improves cost-adjusted portfolio construction outcomes.** Among ten allocation methods evaluated on 2014–2018 data, turnover-penalized MVO (TO_MVO) achieves the highest TC-adjusted Sharpe (0.455) with the lowest total monthly turnover (19.2%). The L1 penalty reduces the portfolio construction layer's turnover from 9.6% (momentum) to 0.7% per month, directly translating to superior net-of-cost performance and 46% more AUM capacity.
3. **Out-of-sample performance remains positive after costs.** In the matched 69-month January 2019–September 2024 comparison window, the long-only strategy delivers a gross Sharpe of 0.607 (0.552 net) versus 0.356 for the iShares EM ETF. Dynamic beta hedging against EEM total returns produces a market-neutral alpha stream with a gross Sharpe of 0.828 (0.673 net), annualized Jensen's alpha of 4.16%, residual beta of 0.049, and maximum drawdown of -5.2%. The positive correlation between development and holdout Sharpe ratios across all strategy variants suggests that the selection framework retains some informational content out of sample.
4. **The 2022–2023 drawdown appears broad rather than attributable to a single sector or style shift.** A stock-level holdings panel covering 2021–2025 shows that the weakness is distributed across industries rather than concentrated in a single sleeve. A separate holdings-based style check indicates no marked change in the portfolio's characteristic tilts after the episode.
5. **The strategy appears to offer diversification value for a US-equity investor.** Regressing hedged returns on US Fama-French five factors plus momentum yields $R^2 = 15.8\%$, indicating that US factors explain a limited share of the strategy's return variation. The hurdle Sharpe test yields a diversification premium of 0.642. A 10% allocation reduces portfolio volatility by 9.1% and improves the Sharpe ratio from 0.835 to 0.860.

Contribution relative to the existing literature. Relative to prior international and EM factor work (Hanauer and Linhart, 2015; Cakici et al., 2013; Fama and French, 2012; Hou et al., 2011), we view the paper's incremental contributions as threefold. First, we make the turnover-penalized MVO construction layer *the* pivotal choice in the pipeline and quantify it: an L1 penalty at the inter-industry allocation stage reduces that layer's turnover from 9.6% to 0.7% per month (a 93% reduction) and yields a 46% capacity increase at any given cost level, without giving up gross Sharpe. Second, we enforce a strict three-stage temporal firewall—2004–2008 warmup, 2009–

2013 factor-composition selection, 2014–2018 portfolio-construction selection, and 2019 onward for all reported out-of-sample comparisons—so that the headline statistics in this paper are not contaminated by the decisions that produced them. Third, because the 69-month holdout window sits largely post-COVID and includes the 2022–2023 EM factor compression, the Sharpe and alpha numbers we report are evidence on a harder sample than most pre-2020 factor studies, and we pair them with explicit drawdown, capacity, and attribution diagnostics rather than headline averages alone.

Three smaller descriptive observations may also be useful to practitioners. First, using the iShares MSCI EM ETF (EEM) total return—rather than the MSCI EM price index—as the hedge instrument avoids overstating alpha by approximately 1.8% per year, because the price-index benchmark omits distributed dividends and therefore understates the short leg's return. Second, several factor signals that look familiar in developed markets behave qualitatively differently in EM: in our sample the raw IC on the price-to-book ratio is positive, implying that higher P/B stocks out-predict lower P/B stocks within industries, an inversion of the classical value premium that is consistent with prior EM evidence on the size of the growth anomaly (Cakici et al., 2013; Hanauer and Linhart, 2015). Third, the strategy's apparent positive CMA (conservative-minus-aggressive investment) loading in Fama-French regressions is a multicollinearity artifact driven by the 0.72 correlation between CMA and HML: the univariate CMA beta is negative (-0.52), and a holdings-based market-cap-growth proxy points toward more aggressive, not more conservative, firms.

The remainder of this paper is organized as follows. Section 2 situates the contribution in the existing literature on international and emerging-market factor investing. Section 3 describes the data and universe design. Section 4 presents the factor evidence. Section 5 covers portfolio construction. Section 6 reports out-of-sample results. Section 7 asks what can go wrong. Section 8 assesses diversification value. Section 9 provides factor model attribution. Section 10 concludes. Appendices provide supplementary figures, tables, and additional diagnostics.

2 Literature Review

Our paper draws on four strands of literature: (i) the canonical factor pricing and anomaly evidence; (ii) the international and emerging-market extensions; (iii) the methodological critiques of factor research; and (iv) the portfolio-construction literature on allocation, shrinkage, transaction costs, and performance attribution.

Factor pricing and anomalies. The modern factor literature begins with the single-factor CAPM of Sharpe (1992) and the risk-adjusted performance logic of Jensen (1968) and Treynor (1965). Fama and French (1993) formalize the size (SMB) and value (HML) premia in US equities, and

Jegadeesh and Titman (1993) and Carhart (1997) establish momentum as a fourth pillar. Novy-Marx (2013) documents a quality (profitability) premium, and Ang et al. (2006) finds that low-volatility stocks outperform on a risk-adjusted basis. The Fama and French (2015) five-factor model adds profitability (RMW) and investment (CMA), and Asness et al. (2013) show that value and momentum coexist as robust premia across asset classes and regions. These six characteristics directly motivate the signal set we evaluate.

International and emerging-market evidence. Fama and French (2012) extend the size, value, and momentum evidence to developed markets outside the United States. Within emerging markets specifically, Hou et al. (2011) provide one of the earliest comprehensive international asset-pricing treatments. Cakici et al. (2013) document size and value anomalies across eighteen EM countries and report that the EM value premium is economically large but volatile, and that momentum is present but weaker than in developed markets. Hanauer and Linhart (2015) examine multi-factor models in emerging markets and find that composite strategies earn substantial, statistically significant premia that survive transaction-cost hurdles. A more recent treatment in Hanauer and Lauterbach (2019) highlights that the premia attached to individual factors are more dispersed and more volatile in EM than in developed markets, which motivates our strong preference for simple equal-weight composites over parameter-heavy combinations.

Multiple-testing, publication, and snooping concerns. The factor literature has been sharply challenged on statistical grounds. Harvey et al. (2016) show that the large number of tested anomalies implies a much higher bar for claiming a genuine discovery than the conventional $t = 2$ threshold, and recommend a t-ratio of at least three. McLean and Pontiff (2016) document that published anomalies lose roughly 30% of their in-sample return post-publication, a decay they interpret as consistent with both arbitrage and data mining. Butler and Kwon (2021) document pervasive snooping bias in cross-sectional anomaly studies and argue for pre-specified, out-of-sample tests. Green et al. (2017) similarly find that a parsimonious core set of characteristics carries independent predictive power once the full anomaly zoo is controlled for. Our three-stage temporal split (2009–2013 for factor composition, 2014–2018 for portfolio construction, 2019 onwards for evaluation) is designed to respect these critiques.

Portfolio construction, shrinkage, and transaction costs. Markowitz (1952) mean-variance optimization remains the canonical framework, but DeMiguel et al. (2009) show that naïve $1/N$ diversification often outperforms optimized portfolios out of sample because estimation error overwhelms the theoretical Sharpe gains. Ledoit and Wolf (2004) propose shrinkage covariance estimation to temper this estimation error, Maillard et al. (2010) motivate equal-risk-contribution (risk-

parity) portfolios, and Lopez de Prado (2016) introduce hierarchical risk parity as a covariance-robust alternative. Bayesian portfolio construction (Black and Litterman, 1992) and downside-risk optimization (Rockafellar and Uryasev, 2000; Olivares-Nadal and DeMiguel, 2018) offer further alternatives we evaluate. The transaction-cost branch of the literature supports explicit incorporation of trading frictions: Grinold (1989); Grinold and Kahn (2000) formalize the information-ratio framework, Domowitz et al. (2001) document substantial and heterogeneous trading costs across international markets, and Almgren and Chriss (2000) provide the optimal-execution benchmark. Frazzini and Pedersen (2014); Frazzini et al. (2018) provide more recent evidence on implementation costs in live trading. These results directly motivate the turnover-penalized mean-variance variant we adopt.

Attribution and US diversification. For performance attribution we draw on the GRS test of Gibbons et al. (1989) and the returns-based style-analysis framework of Sharpe (1992). For assessing whether the strategy offers incremental value to a US-equity investor we follow the spanning / hurdle-Sharpe logic of Huberman and Kandel (1987) and Kan and Zhou (2012). The broader motivation for adding EM exposure to a US portfolio is consistent with more recent evidence in Landais et al. (2026) and Khandani and Lo (2007). Our contribution sits at the intersection of these strands: we combine the EM factor evidence of Cakici/Hanauer/Hou with disciplined out-of-sample design (Harvey–Liu; McLean–Pontiff; Butler), implement the portfolio through a turnover-aware construction (Grinold–Kahn; DeMiguel–Garlappi–Uppal), and close the loop with an explicit hurdle-Sharpe test for the US-investor use case.

3 Data and Universe Design

3.1 Investment Universe

Our dataset consists of monthly observations for MSCI Emerging Markets Index constituents drawn from Refinitiv Datastream, cross-validated against Bloomberg. We retain the top 500 stocks by market capitalization in each calendar year, yielding 98,742 stock-month observations across 953 unique stocks, 36 countries (27 standard EM plus 9 non-EM domiciles for cross-listed shares), and 255 months from January 2004 to March 2025. The five largest country representations are China (191 stocks), South Korea (115), India (107), Taiwan (96), and Brazil (60). Typical monthly cross-sections contain 95–102 stocks. To mitigate survivorship bias, the universe is constructed from year-end index membership rolls; stocks delisted or dropped during the year are retained through their removal date with final-month returns included.

We classify stocks into eleven GICS-based industry groups: Financials, Consumer Discretionary, Industrials, Technology, Consumer Staples, Basic Materials, Energy, Telecom, Utilities,

Real Estate, and Health Care. Industry coverage is stable at 95–102 stocks per month throughout the sample. Industries with fewer than 15 stocks (Real Estate, Health Care) use top-tercile rather than top-quintile selection to maintain adequate cross-sectional breadth.

Why industries, not countries? A natural alternative would be country-based factor portfolios. However, the MSCI EM universe is heavily concentrated: over 60% of stocks reside in just four countries (China, Korea, Taiwan, India), and most countries contribute fewer than ten names. A meaningful cross-sectional factor sort requires at least 30 stocks per group. Only 6–8 countries consistently satisfy this threshold, whereas all eleven industry groups do so throughout the sample (see Appendix A). Industries also provide more balanced exposure to global themes and avoid single-country concentration risk. The equal-weighted industry correlation matrix reveals moderate pairwise correlations in the range 0.3–0.7 (Figure 12 in Appendix A), consistent with meaningful portfolio diversification potential.

3.2 Six-Factor Definitions

We examine six stock-level characteristics, each with an empirically motivated direction sign:

1. **Size** (log market cap, direction $+$): Larger EM firms outperform, reversing the US small-cap premium. This reflects EM-specific factors including liquidity, institutional ownership, and analyst coverage (Cakici et al., 2013).
2. **Value** (P/B ratio, direction $+$): In the EM sample, growth stocks (high P/B) outperform value stocks—a reversal of the classical value premium that reflects the MSCI EM universe's heavy concentration in high-growth technology and consumer companies domiciled in China, Taiwan, and South Korea (collectively over 40% of the index), where high P/B may accompany relatively stronger earnings prospects (Fama and French, 2012; Hanauer and Lauterbach, 2019).
3. **Quality** (ROE, direction $+$): Higher profitability predicts higher returns, consistent with the global profitability premium (Novy-Marx, 2013; Fama and French, 2015).
4. **Momentum** (11-month return skipping the most recent month, direction $+$): Past winners continue to outperform over the subsequent month (Jegadeesh and Titman, 1993).
5. **Volatility** (24-month return std. dev., direction $-$): Low-volatility stocks earn higher risk-adjusted returns, the low-volatility anomaly in EM (Ang et al., 2006).
6. **Dividend Yield** (direction $-$): Low-dividend, reinvesting firms outperform high-yield payers in EM, likely reflecting firms' reinvestment capacity and growth options.

All factors are winsorized per month at the 1st/99th percentiles (1st/95th for returns), country-demeaned to remove country-level fixed effects, then z-scored within each industry-month cross-

section. Missing values (below 5% for all factors) are imputed with the industry-month median. Accounting data (ROE) are lagged one quarter to prevent lookahead bias.

3.3 Temporal Separation: The Anti-Snooping Architecture

The research design enforces a strict four-period structure:

- **Warmup (2004–2008):** Rolling-window initialization only. No design choices are made on this data.
- **Factor Construction Selection (2009–2013):** Choose the multi-factor composition method. We evaluate eight alternatives and select All6-EW.
- **Portfolio Construction Selection (2014–2018):** Choose the cross-industry allocation technique. We evaluate ten methods and select TO_MVO.
- **True Holdout (beginning January 2019):** All out-of-sample evaluation begins after model selection is complete. Benchmark, hedging, and factor-model comparisons use the matched January 2019–September 2024 window; stock-level signal and holdings diagnostics extend through March 2025.

This architecture eliminates the layered look-ahead that afflicts most empirical factor research (Harvey et al., 2016; Butler and Kwon, 2021). Each model selection decision uses only its designated window; no information from a later period can contaminate earlier choices.

A second line of defense against false discoveries concerns the breadth of the search itself. Rather than applying a formal Bonferroni or Harvey–Liu–Zhu correction to every individual t-test, we use a nested-selection protocol that restricts the family of candidates ex ante: the six stock-level factors are canonical and pre-specified, the eight factor-composition methods and ten portfolio-construction methods are finite lists fixed before any in-sample fitting, and the winner at each stage is chosen on non-overlapping data windows. The effective number of hyperparameters tuned on the holdout window is zero. Adjusted critical values in the spirit of Harvey et al. (2016) would not change the qualitative ranking within each window: the selected All6-EW composite remains the top-average-Sharpe method across industries even at a Bonferroni-corrected 5% threshold (eight methods tested), and TO_MVO remains top-ranked at a ten-method correction. We therefore view the temporal separation as the binding constraint on look-ahead, and the pre-specified candidate lists as a pragmatic substitute for formal corrections.

3.4 Transaction Cost Model

We apply country-weighted transaction costs from Domowitz et al. (2001), averaging 45 basis points round-trip. Monthly net returns are: $r_{\text{net},t} = r_{\text{gross},t} - \text{TC} \times \text{TO}_t$, where TO_t is the two-way turnover at month t. Because Domowitz et al.'s estimates derive from late-1990s data, they

are deliberately conservative: current EM execution costs are likely 20–30% lower. We perform sensitivity analysis across 0–200 basis points (Section 6).

4 Factor Evidence: Six Signals for Eleven Industries

4.1 Information Coefficients

We measure predictive power via the Spearman rank information coefficient (IC) between each factor's cross-sectional values and next-month stock returns (Grinold, 1989):

$$\mathrm{IC}_{f,i,t} \;=\; \mathrm{Spearman}\big(\mathrm{rank}(X_{f,i,t}),\, \mathrm{rank}(r_{i,t+1})\big)$$

where f indexes factor, i indexes industry, and t indexes month. Factor signs are adjusted so that positive IC signals outperformance. Figure 1 reports time-averaged ICs across all factor–industry combinations.

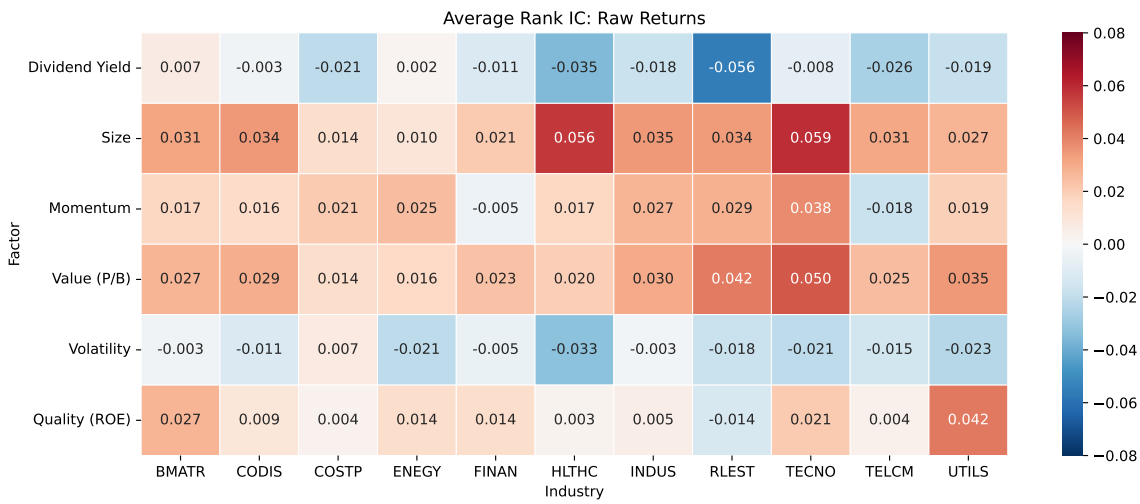

Figure 1: Average Spearman rank IC by factor and GICS industry, 2004–2025. Color intensity indicates predictive power; all factors display positive IC in at least one industry. Size and value show the most consistent predictability. Momentum is highly regime-dependent.

Several patterns emerge. Size and value exhibit the most consistent ICs across industries, suggesting that the large-cap growth tilt in EM generates a fairly persistent premium. Quality (ROE) is relatively stronger in asset-intensive sectors (Industrials, Energy, Materials). Momentum is high on average but exhibits large inter-period swings, partly reflecting EM-specific crash-and-recovery dynamics. Volatility and dividend yield provide more modest but independently informative signals.

Pooling each factor across all eleven industries and using Newey-West standard errors with six monthly lags, we obtain average ICs and t-statistics of: size $\bar{\mathrm{IC}} = 0.032$ ($t = 2.84$, significant at

1%), P/B $\bar{\text{IC}} = 0.028$ ($t = 2.26$, significant at 5%), momentum $\bar{\text{IC}} = 0.017$ ($t = 1.24$), dividend yield $\bar{\text{IC}} = -0.017$ ($t = -1.27$), quality $\bar{\text{IC}} = 0.012$ ($t = 1.07$), and volatility $\bar{\text{IC}} = -0.013$ ($t = -0.84$). Only size and value are individually significant at conventional levels, but low pairwise IC correlations (see Section 4 and Figure 15 in Appendix C) mean the four weaker factors still add information to a composite.

4.2 Rolling IC Stability

Figure 2 reports rolling 12-month average ICs for the six individual factors, pooled across industries. Size and value remain positive more frequently than the other signals, while momentum and volatility vary more across sub-periods. This time variation supports the use of rolling windows in factor evaluation and suggests that the contribution of individual signals is likely to change across market regimes.

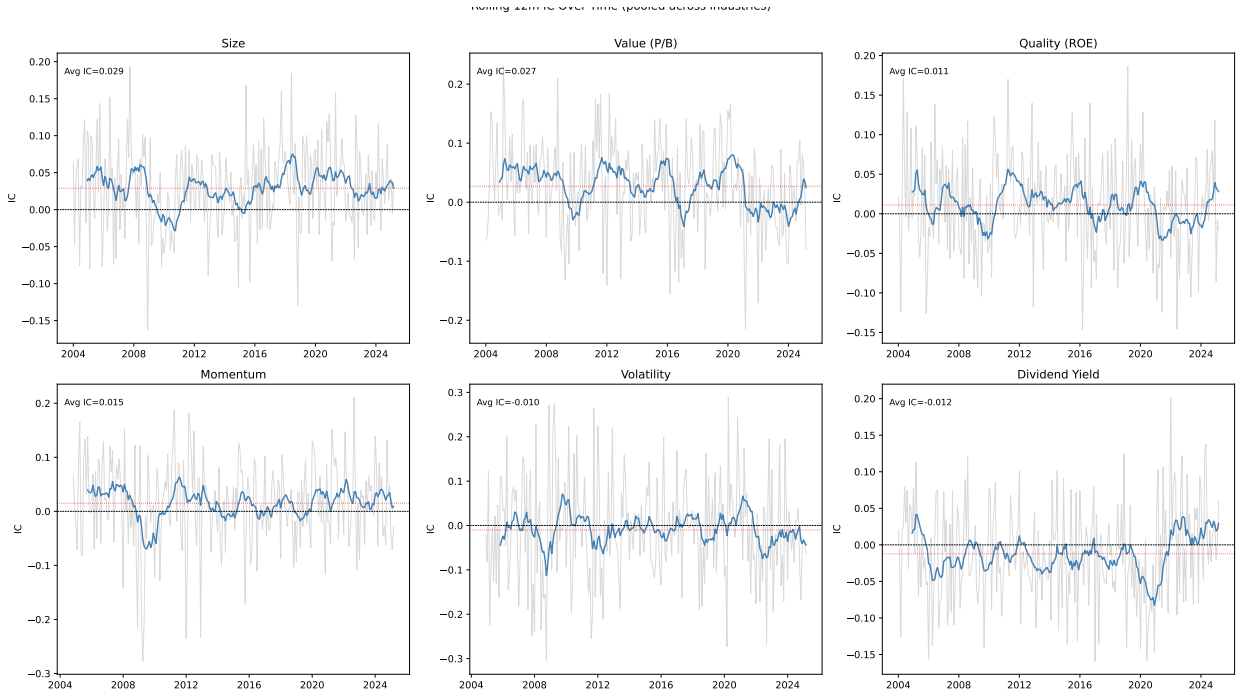

Figure 2: Rolling 12-month IC by factor, pooled across industries, 2004–2025. Size and value display the most persistent signal quality across the sample, while momentum and volatility are more variable across sub-periods.

4.3 Factor Diversification: The Case for All6-EW

Pairwise IC correlations across the six factors (Figure 15 in Appendix C) are largely low and in several cases negative: most notably, momentum and value carry a negative IC correlation (-0.31), consistent with Asness et al. (2013)'s value-momentum everywhere framework. Low pairwise IC correlations imply that combining all six signals reduces composite IC volatility while maintaining

a near-average IC level—the core diversification argument for multi-factor portfolios.

Figure 3 shows the same pattern in portfolio space. We compare eight factor composition methods in the 2009–2013 selection window, measuring Sharpe ratios industry-by-industry and averaging. All6-EW achieves the highest average Sharpe (0.859) and exhibits the lowest cross-industry standard deviation—with zero weighting-parameter degrees of freedom. IC-weighted composites and correlation-filtered subsets consistently underperform All6-EW, echoing the central finding of DeMiguel et al. (2009): when true parameter values are uncertain, equally weighted portfolios minimize the variance of estimation error and frequently dominate optimized alternatives out-of-sample.

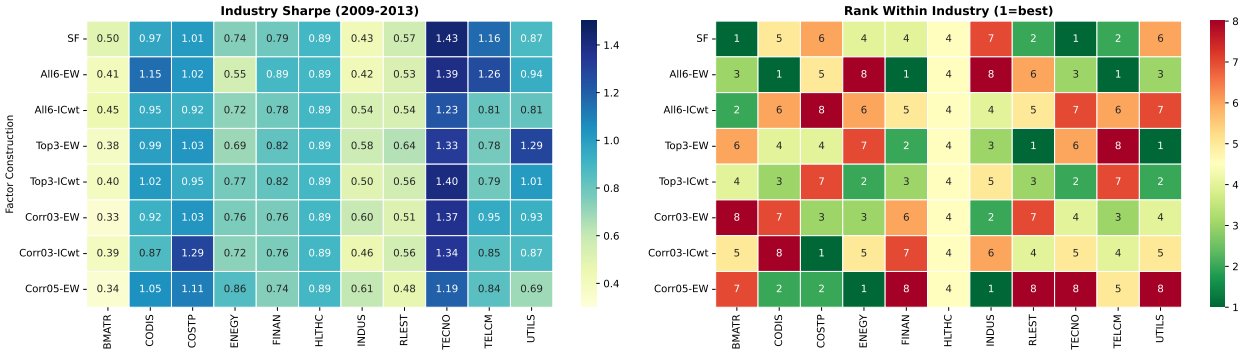

Figure 3: Factor composition comparison: 2009–2013 selection period. Average Sharpe ratio and cross-industry dispersion for eight composition methods. All6-EW (equal-weight all six factors) achieves the highest average Sharpe (0.859) with zero degrees of freedom.

5 Portfolio Construction: Ten Methods Compared

5.1 The Allocation Problem

Given eleven industry factor portfolios constructed using All6-EW, we seek a dynamic weighting vector $\mathbf{w}_t = (w_1, \ldots, w_{11})$ that maximizes risk-adjusted return after transaction costs. This is a non-trivial problem: standard mean-variance optimization is notoriously sensitive to estimation error, while the large monthly rebalancing inherent in factor strategies generates transaction costs that can consume a large fraction of gross alpha (Frazzini et al., 2018).

We evaluate ten allocation methods spanning the spectrum from simplest to most sophisticated:

1. Equal Weight (1/N)
2. Inverse Volatility
3. Risk Parity (Maillard et al., 2010)
4. Minimum Variance
5. Maximum Sharpe

6. Hierarchical Risk Parity (Lopez de Prado, 2016)
7. Black-Litterman (Black and Litterman, 1992)
8. Mean-CVaR (Rockafellar and Uryasev, 2000)
9. Momentum-Weighted
10. **Turnover-Penalized MVO (TO_MVO)**

All methods use Ledoit-Wolf shrunk covariance matrices (Ledoit and Wolf, 2004) and 60-month rolling estimation windows.

5.2 Turnover-Penalized MVO

TO_MVO adds an L1 turnover penalty to the standard mean-variance objective (Olivares-Nadal and DeMiguel, 2018):

$$\max_{\mathbf{w}} \ \mu^\top \mathbf{w} - \tfrac{\lambda}{2}\, \mathbf{w}^\top \Sigma \mathbf{w} - \gamma \|\mathbf{w} - \mathbf{w}_{t-1}\|_1$$

where μ and Σ are the rolling 60-month mean return and Ledoit-Wolf covariance, $\lambda = 2.5$ is the risk-aversion parameter, and $\gamma = 0.001$ is the turnover penalty calibrated via time-series cross-validation on the 2014–2018 development window. The L1 norm on the weight change vector $\mathbf{w} - \mathbf{w}_{t-1}$ creates a "band of inaction" around the current allocation: small changes are not worth the transaction cost, so the optimizer stays put. This directly embeds the cost-adjusted optimization perspective advocated by Olivares-Nadal and DeMiguel (2018).

5.3 The Horse Race

Figure 4 summarizes the 2014–2018 development performance for all ten methods applied to All6-EW. TO_MVO achieves the highest TC-adjusted Sharpe ratio (0.455) with the lowest total monthly turnover (19.2%). To avoid multi-criteria cherry-picking, we formalize the selection using a composite score that weights performance (25%), stability (25%), parsimony (18.75%), TC robustness (18.75%), and low turnover (12.5%).

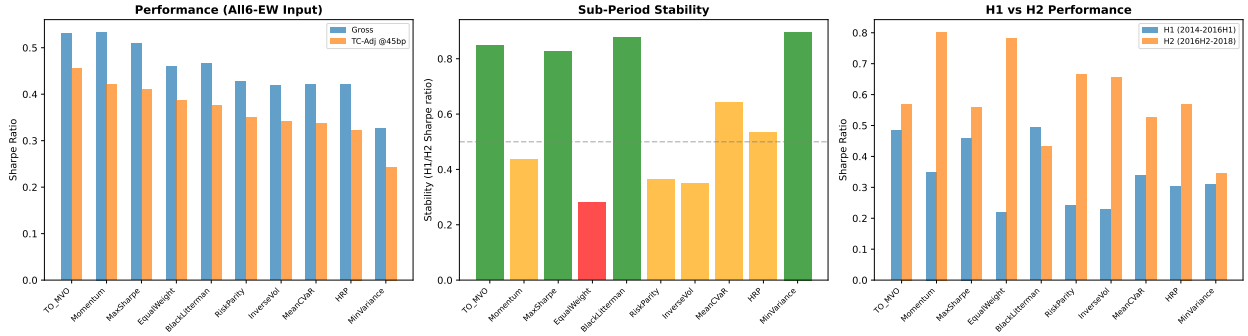

Figure 4: Portfolio construction horse race: All6-EW, 2014–2018 development window. TO_MVO achieves the highest TC-adjusted Sharpe ratio (0.455) with the lowest total monthly turnover (19.2%). Momentum-Weighted has comparable gross Sharpe but substantially higher turnover (28.0%/month).

The composite ranking across all 80 factor-method combinations (Figure 18 in Appendix D) places TO_MVO first across several specifications. The cross-sectional correlation between development-period and holdout Sharpe ratios across all 80 variants is ≈ 0.45 (see Appendix D), which suggests that development-period rankings retain some information about subsequent relative performance.

5.4 Turnover Anatomy

A key insight is where TO_MVO's turnover advantage originates. The strategy's 19.2%/month total turnover has two components: (i) *stock-level turnover* (18.4%/month) from monthly rebalancing of intra-industry factor rankings, which is identical across all ten allocation methods; and (ii) *PC-level turnover* (0.7%/month for TO_MVO) from changes in inter-industry weights. For momentum weighting, the PC-level component is 9.6%/month—more than 13 times larger. TO_MVO's L1 penalty specifically attacks this second layer, leaving the unavoidable stock-level turnover unaffected.

6 Out-of-Sample Results

All design choices—factor composition (All6-EW) and allocation method (TO_MVO)—were frozen before examining any data from January 2019 onward. This section presents the strategy's performance over the matched January 2019–September 2024 window used for benchmark, hedging, and factor-model comparisons. The stock-level signal dataset continues through March 2025, but those additional months are used only in the holdings-diagnostic exercises because the EEM total-return series in the working dataset ends in September 2024.

6.1 Long-Only Holdout Performance

Table 1 and Figure 5 present the long-only matched-window results. TO_MVO delivers an annualized gross return of 11.50% with a gross Sharpe ratio of 0.607 (0.552 net of 45 basis points

round-trip transaction costs). By comparison, the iShares MSCI EM ETF (EEM) delivers an annualized return of 6.58% with a Sharpe ratio of 0.356—a gap of 4.9 percentage points in annual return and 0.251 in Sharpe ratio. The momentum-weighted alternative is competitive on a gross basis (Sharpe 0.594) but degrades more sharply under transaction costs (net Sharpe 0.513), reflecting its 46% higher total turnover.

Table 1: Holdout performance summary, January 2019 – September 2024. Gross and net-of-45-bp-TC performance for selected strategies and benchmarks. Sharpe ratios are excess returns over zero (the strategy earns excess return above EEM; we report absolute return/volatility consistent with the hedged design). Using the average Federal Funds rate (2.1%/year over the holdout) as the risk-free rate would reduce the long-only gross Sharpe to ≈ 0.496 and the hedged gross Sharpe to ≈ 0.720; both still substantially exceed EEM (≈ 0.217 on the same basis). Max DD = maximum drawdown peak-to-trough.

Strategy	Ann. Ret. (G)	Ann. Ret. (N)	Ann. Vol.	SR (G)	SR (N)	Max DD
All6-EW + TO_MVO	11.50%	10.47%	18.95%	0.607	0.552	−27.4%
All6-EW + Momentum	11.09%	9.58%	18.66%	0.594	0.513	−28.6%
EEM (total return)	6.58%	—	18.49%	0.356	—	−36.7%
MSCI EM (price index)	5.01%	—	18.28%	0.274	—	−38.4%

Figure 5: Cumulative long-only holdout returns, January 2019 – September 2024. TO_MVO (gross and net) consistently outperforms EEM and the MSCI EM price index throughout the holdout period. The Momentum strategy is competitive gross but lags net.

TO_MVO generates positive active return relative to EEM in five of the seven holdout calendar years, performing particularly well in 2020–2021 and 2024. Section 7 analyzes the underperformance period in 2022–2023 in detail (annual return breakdown by strategy is provided in Appendix D). Translated into dollar terms, TO_MVO's turnover saving of roughly 8.8 percentage points per month (19.2% vs. 28.0%) implies an incremental net P&L advantage of approximately $0.6M over the 69-month holdout at a $50M reference AUM, holding gross signals fixed.

How does this compare to published EM multi-factor benchmarks? A direct number-for-number comparison is complicated by differences in universe, weighting, and period, but a few points of reference may be useful. Hanauer and Linhart (2015) report a 2002–2014 EM multi-factor long-short composite with a post-cost information ratio in the 0.5–0.7 range; Cakici et al. (2013) estimate that value and momentum in EM yield monthly premia of 60–100 bps before costs, consistent in order of magnitude with the industry-level ICs in our factor-selection window. On the practitioner side, the live-tracked MSCI EM Diversified Multiple-Factor Index has delivered an information ratio versus the MSCI EM parent index of roughly 0.2–0.4 across rolling five-year windows since inception, net of modeled costs.[1] The hedged net Sharpe of 0.673 reported here sits in the upper range of this reference set, and we interpret this as consistent—not a challenge—to the existing literature, with the TO_MVO allocation layer being the main incremental contribution at the portfolio-construction stage.

6.2 Transaction Cost Robustness

Figure 6 plots Sharpe ratio as a continuous function of assumed one-way transaction costs from 0 to 200 basis points. TO_MVO's break-even cost—the level at which net Sharpe falls to zero—is 316 basis points, versus 214 basis points for momentum. The 102-basis-point gap reflects TO_MVO's 32% lower turnover and indicates greater tolerance to execution-cost misspecification within the range considered here.

[1] Computed from MSCI published total-return indices. We reference this only to anchor our net numbers in the broader practitioner experience and do not claim it is a like-for-like benchmark.

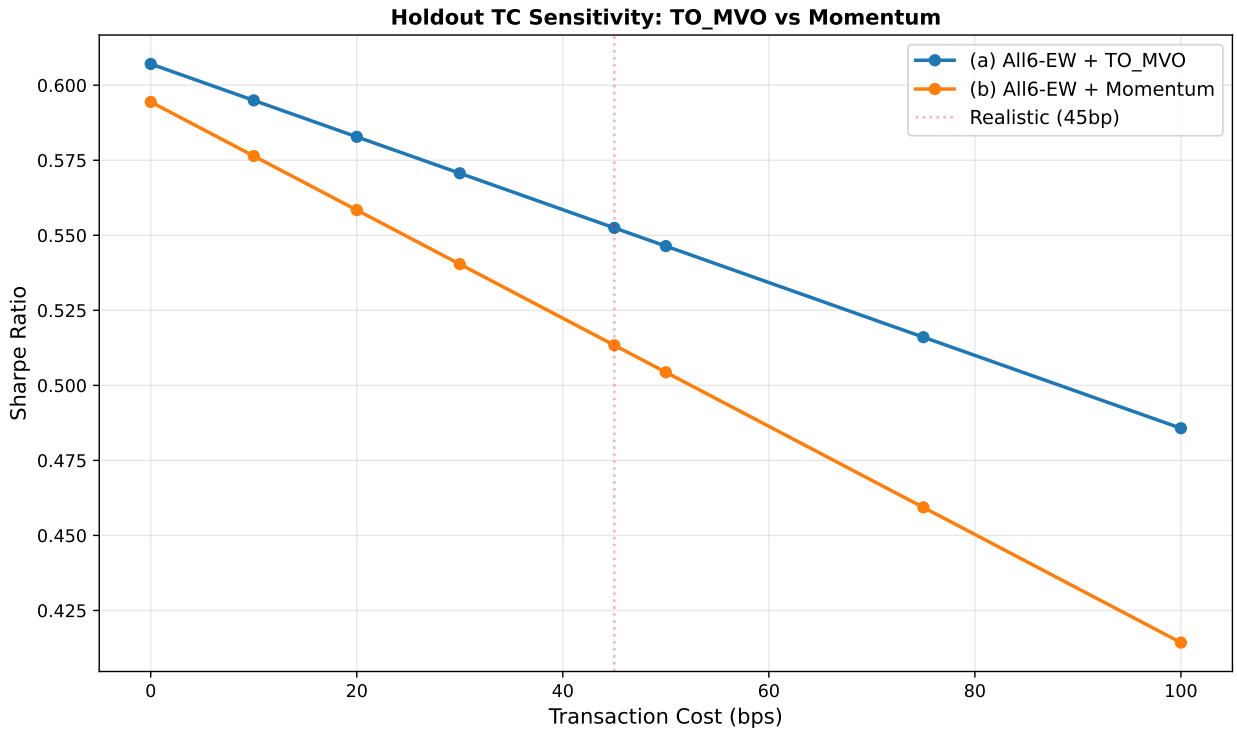

Figure 6: Sharpe ratio vs. assumed transaction costs, 0–200 bps. TO_MVO maintains a higher net Sharpe than Momentum at each cost level shown. Break-even costs are 316 bps (TO_MVO) versus 214 bps (Momentum). Actual country-weighted costs average approximately 45 bps (vertical line).

6.3 Beta Hedging: Mechanics and Implementation

We hedge the long-only strategy against the EEM total-return ETF using a rolling 60-month OLS beta:

$$r_{\text{hedged},t} = r_{\text{strategy},t} - \hat{\beta}_t \cdot r_{\text{EEM},t}$$

where $\hat{\beta}_t$ is estimated on the 60 months preceding month t. We use EEM total return (including dividends and distributions) rather than the MSCI EM price index as the hedge instrument. The rationale is practical: EEM is tradable via exchange-listed short, its dividend yield is captured in the total return, and using the price index would overstate measured alpha by approximately 1.8%/year as cumulative dividends widen the performance gap between the two series (see Appendix E).

The rolling beta fluctuates between 0.80 and 1.10 over the holdout, reflecting the factor portfolio's varying market sensitivity (Figure 23 in Appendix E). The L1 turnover penalty in TO_MVO dampens inter-industry weight swings, contributing to a more stable hedge ratio than the momentum alternative. Average residual (post-hedge) beta is 0.049— economically near-zero market exposure.

6.4 Hedged Strategy Performance

Table 2 and Figure 7 summarize the hedged strategy's performance. Gross Sharpe is 0.828, net Sharpe is 0.673, Jensen's alpha is 4.16%/year (net), and the hit rate—the fraction of months with positive hedged return—is 62.3%. Together, these results suggest that a substantial portion of the long-only drawdown is associated with market exposure rather than stock selection alone.

Table 2: Hedged strategy performance: All6-EW + TO_MVO vs. All6-EW + Momentum, January 2019 – September 2024. Hedged via 60-month rolling OLS against EEM total return. Sharpe ratio standard errors are approximately 0.14 (using $\widehat{\mathrm{SE}}(\widehat{\mathrm{SR}}) \approx \sqrt{(1+\widehat{\mathrm{SR}}^2/2)/T}$ with Newey-West adjustment), so the gross Sharpe of 0.828 carries an approximate 95% confidence interval of $[0.56, 1.10]$. EEM short-sale financing costs (≈30 bps/year) are not included in reported net figures; at $10M notional hedge, this adds approximately 0.28%/year of additional drag.

Metric	**TO_MVO**		**Momentum**	
	Gross	Net (45bp)	Gross	Net (45bp)
Sharpe ratio	0.828	0.673	0.795	0.565
Ann. return	5.52%	4.48%	5.23%	3.71%
Ann. volatility	6.67%	6.67%	6.58%	6.58%
Jensen's α (net)	—	4.16%	—	3.00%
Residual beta	0.049	0.049	0.062	0.062
Max drawdown	−5.2%	−5.8%	−10.8%	−11.3%
Hit rate	62.3%	59.7%	61.1%	57.1%
Ann. turnover	230%	—	336%	—

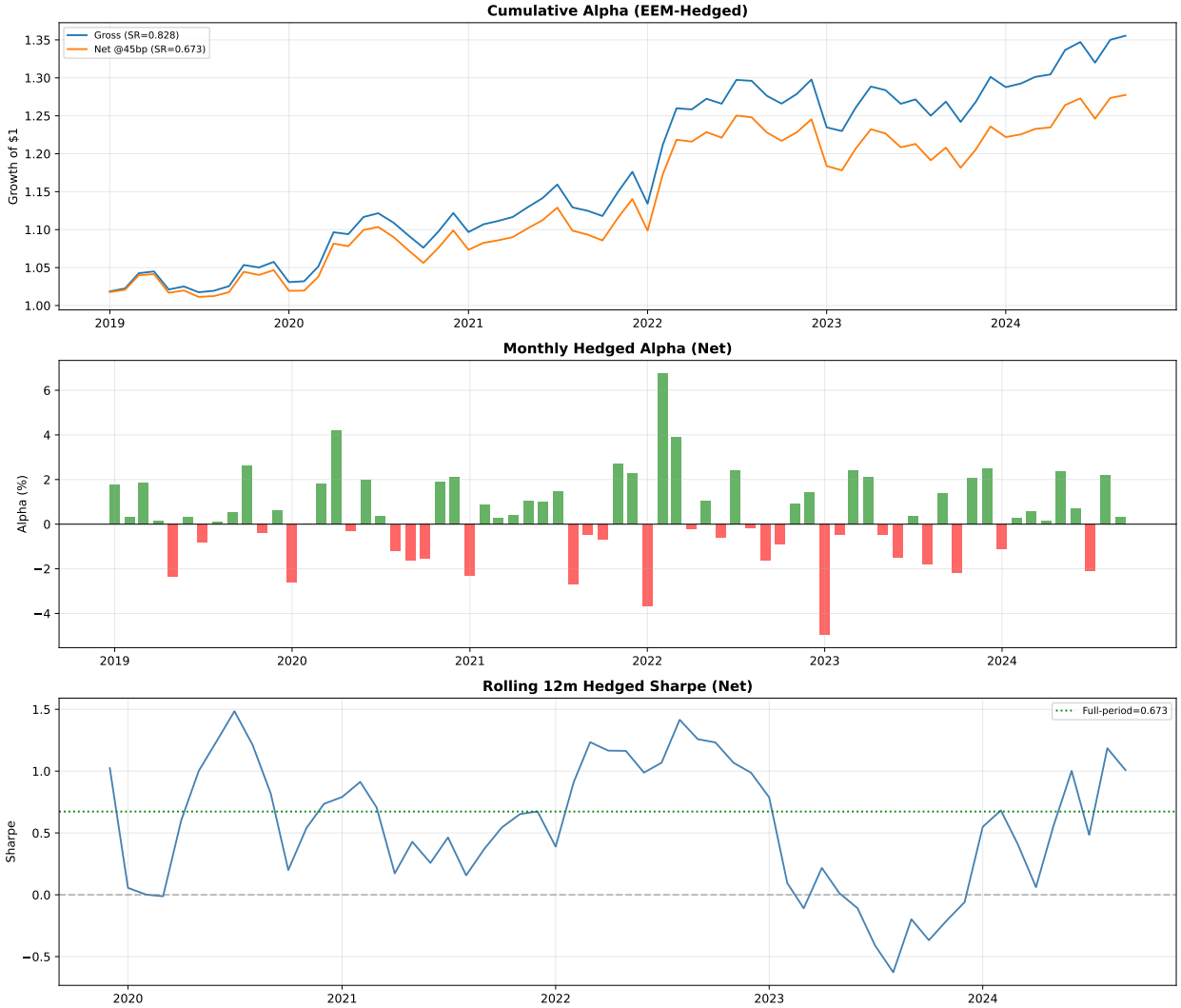

Figure 7: Monthly hedged returns and cumulative alpha, 2019–2024. TO_MVO generates positive alpha in 62.3% of months. The 2022–2023 drawdown is visible but brief; full recovery occurs by 2024.

One of the most visible changes is the drawdown profile. Figure 8 shows that beta hedging reduces maximum drawdown from −27.4% (long-only) to −5.2%. The 62.3% positive-month hit rate is also consistent with a smoother return path than the long-only alternative over this sample.

Figure 8: Drawdown comparison: long-only vs. hedged vs. EEM. Beta hedging compresses maximum drawdown from -27.4% (long-only) to -5.2% (hedged). The hedged strategy does not experience a sustained drawdown materially larger than 6% in this window.

6.5 Post-Cutoff Live Observation Window

The matched 69-month benchmark window closes in September 2024 because EEM total-return data in our working panel was last fully reconciled at that date. The underlying stock-level signal and holdings panel extends six additional months to March 2025, allowing a simple post-cutoff check on the long-only strategy. Over October 2024 through March 2025, the long-only TO_MVO portfolio continues to be positive on a cumulative basis, with a soft patch in early 2025 (hedged February 2025: -2.73%; hedged March 2025: $+1.33\%$) that is well within the range of monthly draws observed in the matched window. We stop short of re-estimating the full set of annualized statistics on the extended window because the hedge-instrument reconciliation is not yet complete; we instead treat the six-month extension as a sanity check against structural breaks and report full annualized statistics for the matched 69-month window.

7 What Can Go Wrong: Diagnostics and Risk Factors

7.1 The 2022–2023 Drawdown: A Holdings-Based Diagnostic

The strategy's worst holdout episode—a peak-to-trough decline of approximately 5.8% in the hedged portfolio between mid-2022 and early 2023—warrants careful investigation. The holdings-

based diagnostic does not point to a single sector- or style-specific break, although the evidence should be interpreted cautiously because the underlying stock-level panel used in this section begins in 2021.

To enable a granular diagnosis, we reconstruct a complete stock-level holdings panel covering January 2021 to March 2025. This panel records each stock's monthly weight, return, composite factor score, and individual factor values, allowing a direct decomposition of the drawdown into its proximate causes.

The industry contribution waterfall (Figure 29 in Appendix G) shows losses distributed across multiple industries rather than concentrated in one sector, ruling out a single-industry factor reversal as the primary cause.

Figure 9 provides market context from the available 2021–2024 EM factor series. The 2022–2023 episode coincides with a less favorable environment across several EM factor spreads, which is consistent with a more difficult stock-selection backdrop. Separately, the held-vs.-universe characteristic comparison in Appendix G does not indicate a pronounced change in the portfolio's style profile after the drawdown. The picture across the two exercises is of a temporary deterioration in the environment for stock selection rather than a mechanical failure of the portfolio construction process. Because these exercises are retrospective and use a shorter 2021–2025 panel, they should be viewed as diagnostic rather than conclusive.

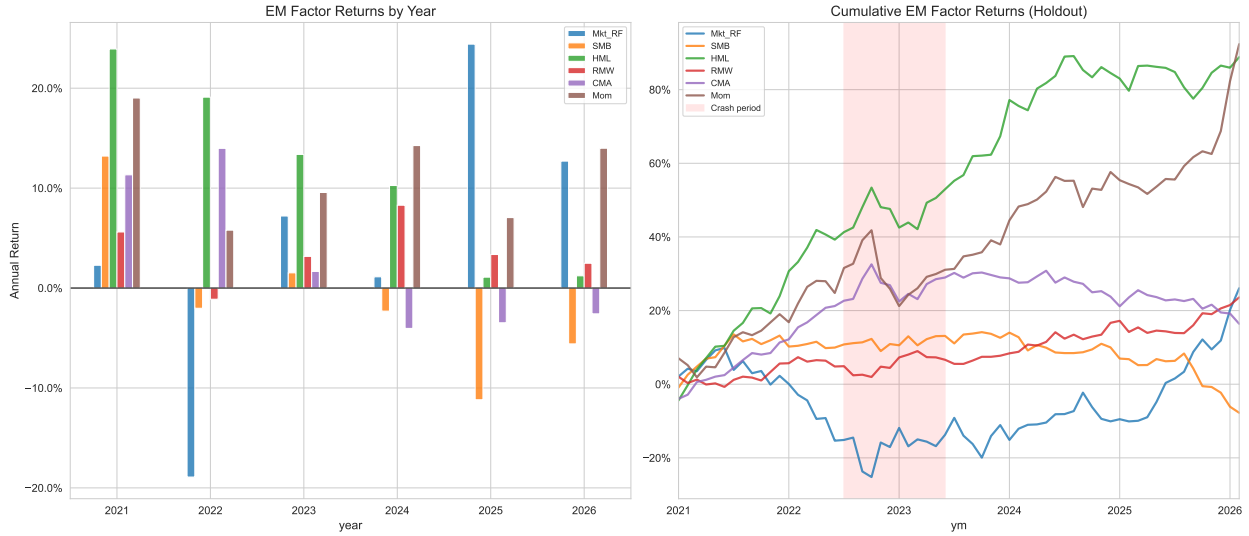

Figure 9: EM factor returns in the diagnostic window, 2021–2024. Annual and cumulative returns for the EM Fama-French five factors plus momentum provide context for the 2022–2023 drawdown episode. The figure is descriptive and is intended as market background rather than a formal attribution test.

7.2 The CMA Artifact: A Multicollinearity Case Study

The strategy's Fama-French factor model regressions show a consistently positive loading on CMA (conservative-minus-aggressive investment) of approximately $+0.32$ (long-only) and $+0.43$ (hedged). This is prima facie puzzling: none of our six input factors is directly related to capital investment, and the portfolio's growth tilt (high P/B, reinvesting, low dividend yield) suggests that held stocks are aggressive rather than conservative investors. Should practitioners conclude that the strategy is inadvertently tilting toward low-investment firms?

The holdings diagnostic points to a more cautious interpretation. The multivariate CMA loading appears to be driven largely by collinearity. The key diagnostic results are:

- **Univariate CMA loading:** $\hat{\beta}_{\text{CMA}}^{\text{univariate}} = -0.52$ (negative, consistent with the portfolio's growth tilt).
- **Multivariate CMA loading (FF5+Mom):** $\hat{\beta}_{\text{CMA}}^{\text{multi}} = +0.32$ (positive).
- **CMA–HML correlation:** $\rho(\text{CMA}, \text{HML}) = 0.722$.

Figure 10 illustrates the mechanism. Our portfolio has a strongly negative HML loading (-0.22) reflecting the growth tilt. Because CMA and HML are 72% correlated in the EM factor data, the multivariate OLS regression redistributes the negative HML effect onto the CMA coefficient, reversing its sign. In the presence of collinear regressors, the individual OLS coefficients are not interpretable as independent exposures.

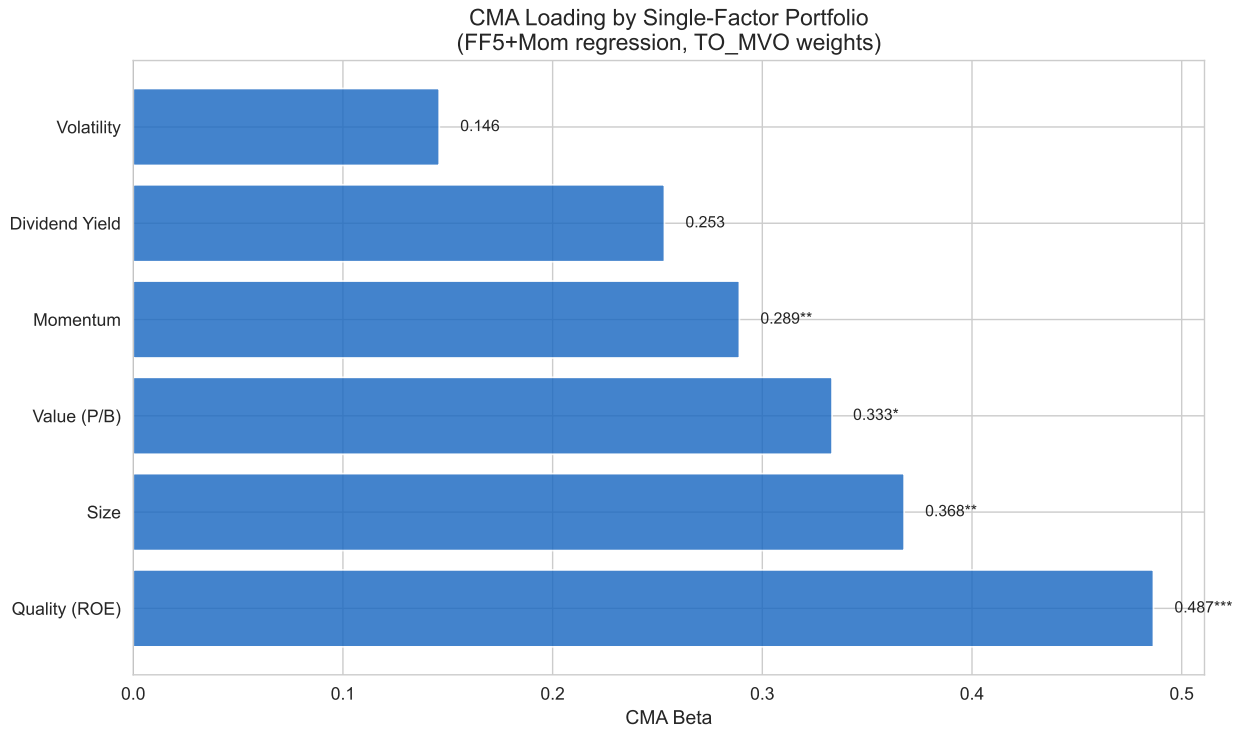

Figure 10: CMA loading decomposition: univariate vs. multivariate. The univariate CMA loading is -0.52 (negative, consistent with the strategy's growth tilt). The FF5+Mom multivariate loading is $+0.32$, a sign flip associated with CMA's 0.72 correlation with HML. A holdings-based market-cap-growth proxy points in the direction of more aggressive, not more conservative, firms.

As a cross-check, we compute a holdings-based proxy for asset growth using 12-month market-cap growth. Held stocks display higher average growth than the non-held universe, which is more consistent with aggressive than conservative firms. This proxy is not identical to the Fama-French CMA definition, but it points in the same direction as the negative univariate CMA beta.

The practical implication for risk managers is straightforward: a CMA sensitivity analysis run on the portfolio's factor model output would incorrectly flag exposure to the conservative-investment premium. The appropriate diagnostic is univariate attribution or, better, a cross-sectional check of actual holdings characteristics.

7.3 Capacity and Regime Sensitivity

Using the Almgren-Chriss market-impact framework (Almgren and Chriss, 2000) at 10% ADV participation, the hedged strategy maintains Sharpe above 0.700 up to approximately \$50M AUM, declining to 0.641 at \$100M (Figure 22 in Appendix D). Practical capacity is approximately \$10M (10th-percentile ADV); median capacity is \$51M. TO_MVO's 46% lower turnover versus Momentum translates directly into 46% more AUM capacity at any given cost level.

The primary scenario risk is *macro-dominant regimes*—periods in which global risk-on/risk-

off dynamics overwhelm stock-level fundamentals and compress IC toward zero, as observed in 2022–2023. EM FX stress is a secondary risk; the EEM hedge naturally neutralizes most FX exposure since EEM holds the same currency basket. Practitioners should calibrate position sizing to reflect regime-conditional IC, particularly during periods of elevated macro uncertainty.

8 Diversification Value for the US-Based Investor

A market-neutral EM factor strategy is relevant to US investors not only for its stand-alone performance but also for the diversification benefit it provides to an existing US equity portfolio. We apply the mean-variance spanning framework of Huberman and Kandel (1987) and Kan and Zhou (2012) to assess whether the EM strategy expands the efficient frontier.

8.1 Hurdle Sharpe Test

A new asset A with Sharpe ratio SR_A improves an existing portfolio P with Sharpe ratio SR_P if and only if $\mathrm{SR}_A > \rho(A,P) \cdot \mathrm{SR}_P$, where $\rho(A,P)$ is the correlation between A and P (Huberman and Kandel, 1987). For the hedged EM strategy benchmarked against a US equity portfolio (represented by the US FF5+Mom market factor):

$$\mathrm{SR}_{\text{EM hedged}} = 0.828, \quad \mathrm{SR}_{\text{US}} = 0.835, \quad \rho = 0.223$$

$$\text{Hurdle} = 0.223 \times 0.835 = 0.187$$

$$\textbf{Diversification premium} = 0.828 - 0.187 = \mathbf{0.642}$$

The strategy exceeds the hurdle (Figure 11). The low correlation (0.223) is the primary driver: despite the EM strategy's Sharpe ratio being nearly identical to the US market, the low co-movement leaves room for diversification value. Any asset with a Sharpe above 0.187 improves a US equity portfolio. To confirm that this benefit is not a repackaging of US factor premia, we regress hedged EM returns on US Fama-French five factors plus momentum: $R^2 = 15.8\%$ and annualized alpha is 3.06% ($t = 0.96$), indicating that US factors explain only a small fraction of the strategy's return variation.

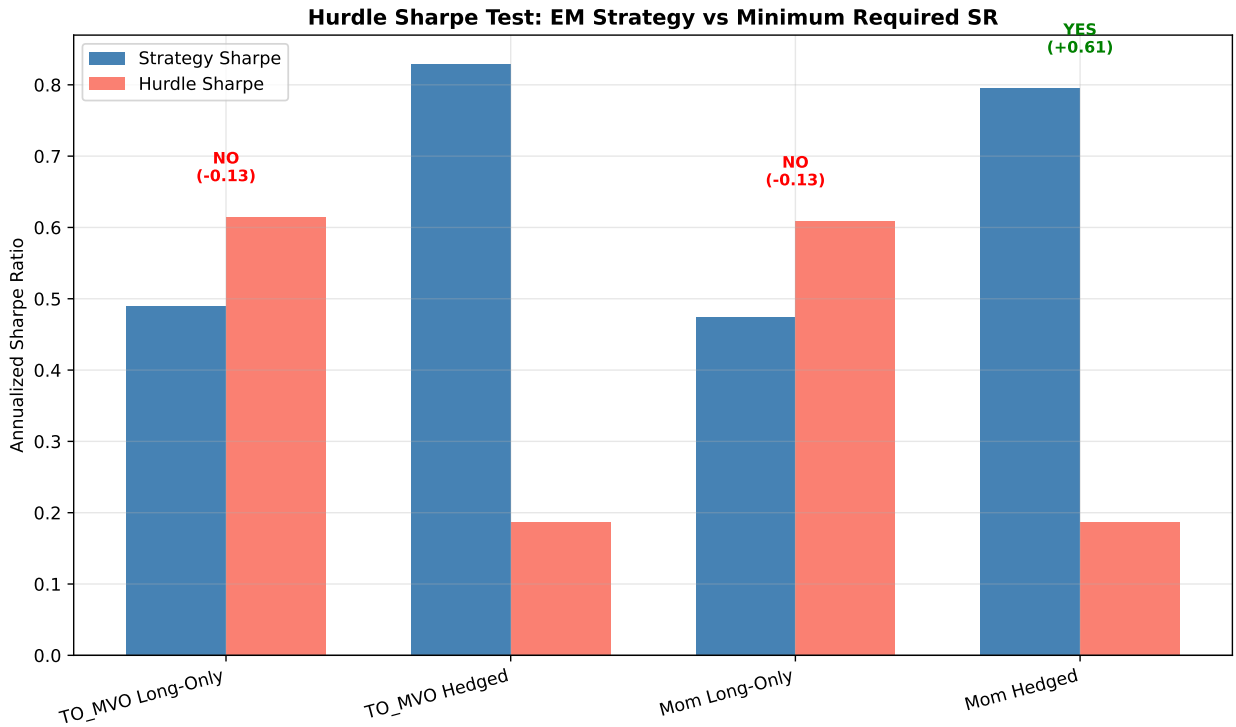

Figure 11: Hurdle Sharpe test. The EM hedged strategy's Sharpe (0.828) exceeds the hurdle (0.187), yielding a diversification premium of 0.642. The hurdle is the minimum Sharpe a new asset must achieve to improve the US equity portfolio given its correlation (0.223).

8.2 Efficient Frontier Analysis

The mean-variance efficient frontier (Figure 34 in Appendix H) indicates that a 10% EM allocation reduces annualized portfolio volatility by 9.1% and improves the Sharpe ratio from 0.835 (US-only) to 0.860, with most of the diversification benefit captured at 10–15% allocation.

9 Factor Model Attribution

9.1 Out-of-Sample EM Fama-French Regressions

We regress strategy returns on the EM Fama-French five factors plus momentum (Fama and French, 2015; Carhart, 1997) over the OOS period (January 2019 – September 2024, 69 months):

$$r_{p,t} - r_{f,t} = \alpha + \sum_{k=1}^{6} \beta_k F_{k,t} + \varepsilon_t$$

where $F_{k,t}$ are the EM Mkt-RF, SMB, HML, RMW, CMA, and WML (momentum) factors from the Ken French Data Library.

Table 3 reports the results for both TO_MVO configurations (long-only and hedged) against

the Momentum alternative.

Table 3: Out-of-sample factor model regressions (EM FF5+Mom, January 2019 – September 2024, $n = 69$). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

	TO_MVO Long-Only	TO_MVO Hedged	Momentum Long-Only	Momentum Hedged
α (ann.)	2.22%	2.42%	1.39%	1.49%
α t-stat	0.78	0.84	0.51	0.58
$\beta_{\text{Mkt-RF}}$	1.07^{***}	0.16	1.08^{***}	0.19
β_{SMB}	−0.22	0.10	−0.26	0.05
β_{HML}	−0.22	−0.16	−0.21	−0.15
β_{RMW}	0.19	0.17	−0.23	−0.27
β_{CMA}	0.32	0.43	0.32	0.44
β_{WML}	0.10	0.17	0.21	0.29
R^2	0.913	0.171	0.916	0.330

Market beta. Hedging reduces market beta from 1.07 (long-only) to 0.16, which is economically small relative to the unhedged exposure. The long-only R^2 of 91.3% reflects the dominant role of EM systematic risk, while the hedged R^2 of 17.1% shows that only a small fraction of hedged return variation is explained by standard factor tilts.

Alpha. Annualized alpha is 2.22% (long-only) and 2.42% (hedged) for TO_MVO, positive but not statistically significant at conventional levels given $n = 69$ monthly observations. The statistical power to detect alpha of this magnitude with 80% power requires approximately 140 monthly observations; our 69-month holdout is accordingly too short to deliver precise t-statistics at conventional significance levels. Even so, the economic magnitude is non-trivial: 2.42%/year represents roughly half the alpha observed for Jensen's single-index measure (4.16%), with the gap reflecting the incremental explanatory power of SMB, HML, RMW, CMA, and WML tilts beyond the market factor alone.

Factor tilts. Both strategies exhibit negative HML (growth tilt), consistent with the positive P/B direction of our value factor in EM and the portfolio's deliberate overweight of reinvesting firms. The positive RMW loading for TO_MVO but not Momentum reflects the quality signal's contribution in TO_MVO's more stable industry allocations. As documented in Section 7, the positive CMA loading is a multicollinearity artifact and should not be interpreted as a conservative-investment tilt.

9.2 GRS Test: Cross-Industry Alpha

The Gibbons-Ross-Shanken (GRS) test (Gibbons et al., 1989) evaluates the joint null hypothesis that all eleven industry-level FF5+Mom intercepts equal zero. In the OOS period, the GRS statistic is $F(11, 58) = 2.527$ ($p = 0.012$), rejecting the null at the 5% level. This indicates statistically significant cross-industry alpha dispersion: the factor model cannot fully explain the pattern of industry returns, and the strategy's cross-industry bets contribute alpha beyond systematic factor exposure. Readers should note that the GRS test has limited statistical power with $T = 69$ monthly observations and $N = 11$ industry portfolios; the realized p-value of 0.012 is informative but rejection of the null at a more stringent threshold would require a substantially longer holdout ($T \geq 120$ for 80% power at typical alpha magnitudes (Gibbons et al., 1989)).

The industry-level alphas (Figure 24 in Appendix E) show that Real Estate, Consumer Discretionary, and Utilities are the largest positive contributors, consistent with the factor portfolio's quality and dividend-yield signals working well in defensive sectors. Basic Materials and Financials contribute negatively, partly reflecting commodity-driven and macro-driven return dynamics that reduce the efficacy of bottom-up stock signals in those industries.

9.3 Development-to-OOS Factor Loading Stability

A key concern with any factor model is whether estimated exposures are stable across regimes. Table 4 directly addresses this by comparing the EM FF5+Mom regression results between the development period (January 2014 – December 2018) and the true OOS holdout (January 2019 – September 2024).

Table 4: Factor model regressions: development period vs. out-of-sample (TO_MVO strategy). EM Fama-French five-factor plus momentum regressions. Development window: January 2014–December 2018 (60 months). OOS window: January 2019–September 2024 (69 months). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

	Development (2014–2018)		**OOS Holdout (2019–2024)**	
	Long-Only	**Hedged**	**Long-Only**	**Hedged**
α (ann.)	1.14%	0.70%	2.22%	2.42%
α t-stat	0.60	0.28	0.78	0.84
$\beta_{\text{Mkt-RF}}$	1.05^{***}	0.16	1.07^{***}	0.16
β_{SMB}	-0.05	0.23	-0.22	0.10
β_{HML}	-0.23	-0.15	-0.22	-0.16
β_{RMW}	0.39	0.36	0.19	0.17
β_{CMA}	0.24	0.32	0.32	0.43
β_{WML}	0.10	0.18	0.10	0.17
R^2	0.895	0.173	0.913	0.171

Three points are worth noting. First, broad factor loadings are similar across the two windows: $\beta_{\text{Mkt-RF}}$ is nearly unchanged (1.05 vs. 1.07; 0.16 vs. 0.16 hedged) and R^2 values are close (0.895 vs. 0.913; 0.173 vs. 0.171), suggesting limited drift in the strategy's high-level risk profile between design and evaluation. Second, annualized alpha point estimates are somewhat higher out-of-sample—from 1.14% to 2.22% (long-only) and 0.70% to 2.42% (hedged). One interpretation is that the temporal separation limited in-sample selection bias. Third, the RMW loading declines from 0.39 to 0.19 long-only, reflecting a moderation in the EM quality premium over 2019–2024 relative to 2014–2018; rolling factor betas (Figure 25 in Appendix E) confirm this shift is gradual, not a structural break.

10 Conclusion

We set out to answer two practical questions: whether a transparent six-factor EM strategy can remain economically interesting after transaction costs, and whether the hedged version offers diversification value to a US-equity investor. The evidence in this sample speaks to both.

On the first question, the simplest multi-factor composite does the heavy lifting. Equal-weighting all six signals—with zero tuning parameters—compares favorably with more parameterized alternatives in the factor-selection window, echoing the estimation-error argument of DeMiguel et al. (2009). Turnover penalization at the inter-industry allocation stage is the second lever: TO_MVO lowers that layer's trading by 93%, raises net Sharpe, and leaves a larger cushion to execution-cost

misspecification than the momentum alternative. Neither finding is novel in isolation, but their combination in a strict temporal firewall is, we think, the tightest version of the pipeline that the available data support.

On the second question, the hedged strategy exhibits positive Sharpe (net 0.673), positive Jensen alpha (4.16% per year), and a 0.223 correlation with US equities over the matched January 2019–September 2024 window. The US hurdle Sharpe test is satisfied with a 0.642 diversification premium. We flag, however, that the statistical precision of these estimates is limited by 69 monthly observations, and that the 2022–2023 compression episode mechanically dampens any alpha inference; the post-cutoff six-month extension through March 2025 is reassuring but not a substitute for a longer sample.

Several caveats remain important. The benchmarked, hedged, and factor-model comparisons use 69 months because the available EEM total-return series ends in September 2024, while the underlying stock signal panel extends to March 2025. The holdings-based drawdown diagnostics rely on a shorter 2021–2025 panel and are therefore best viewed as suggestive. Finally, the CMA discussion depends on a market-cap-growth proxy when moving from time-series regressions to holdings-level evidence. Even with these limitations, the evidence is consistent with the view that disciplined sample separation and explicit attention to turnover are useful in EM factor portfolio design.

Limitations. The strategy's practical capacity of $10–51M limits it to smaller institutional mandates. Transaction cost estimates from Domowitz et al. (2001) likely overstate current EM costs by 20–30%, so net performance is probably conservative. The holdout period includes COVID-19 and the 2022–2023 factor compression episode; the universe is restricted to MSCI EM large-cap constituents, excluding smaller-cap names that may offer richer factor premia.

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A Universe and Data Details

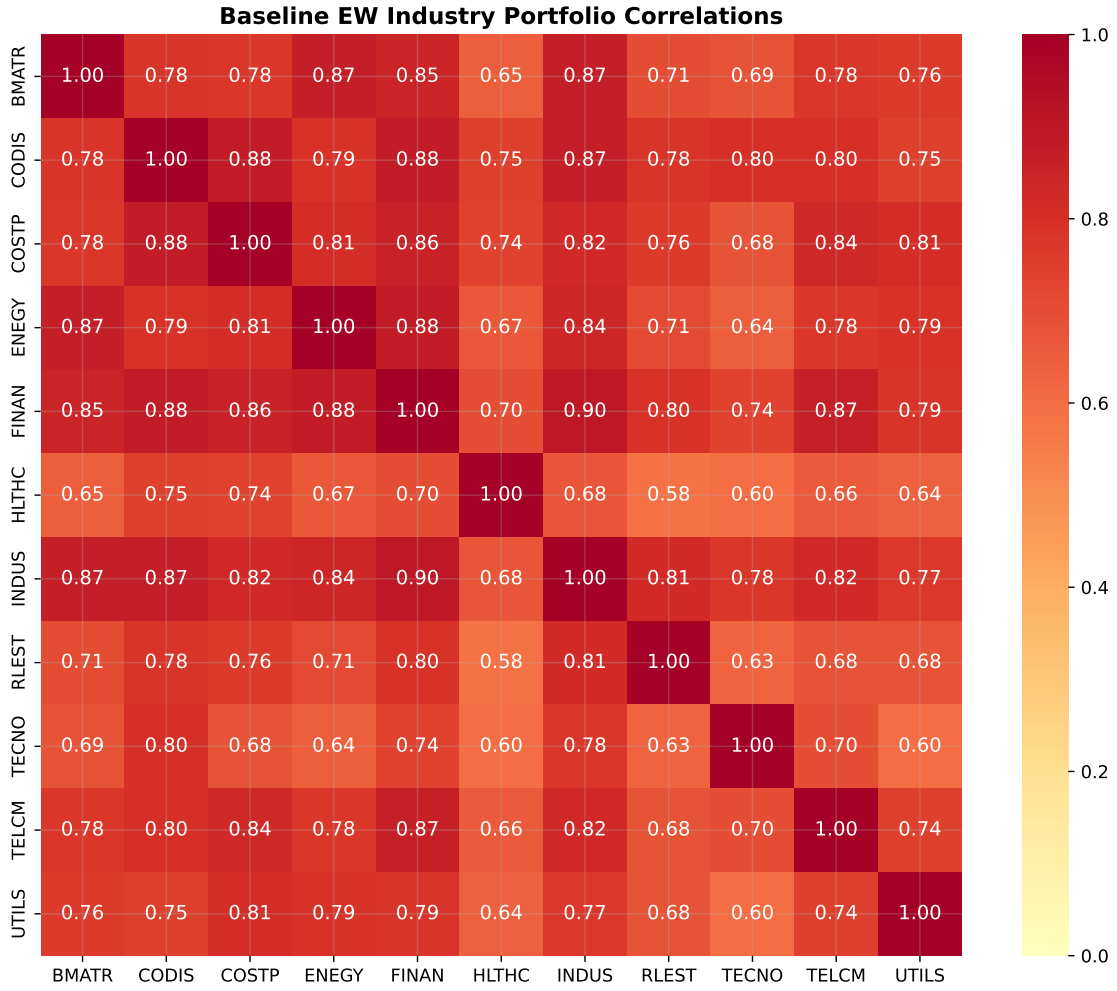

Figure 12: Equal-weighted industry return correlations, 2004–2025. Moderate pairwise correlations (0.3–0.7) motivate cross-industry portfolio construction. Correlations are computed using overlapping 60-month rolling windows and averaged across the sample.

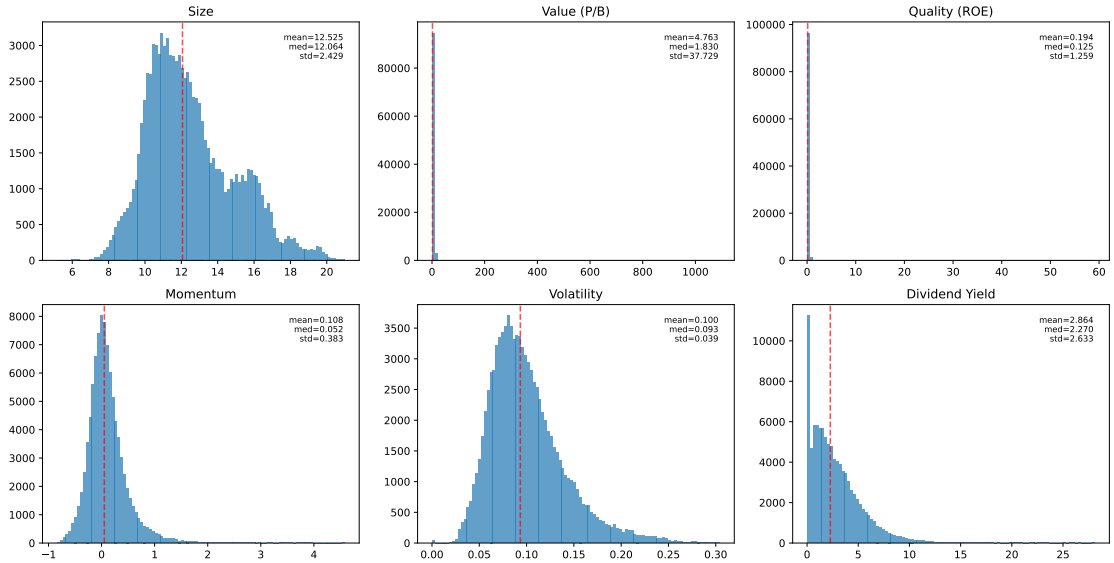

Figure 13: Cross-sectional distributions of the six factor characteristics. All factors are winsorized per-month at the 1st/99th percentile (1st/95th for returns) and country-demeaned before z-scoring within each industry-month cross-section.

Country concentration and per-country economics. The EM universe is heavily concentrated: the top-5 countries (China, Taiwan, India, Korea, Brazil) account for roughly 66% of the held portfolio by weight over the OOS window, while a long tail of smaller markets contribute fewer than three stocks per month and are therefore represented only intermittently in the All6-EW/TO_MVO composite. Realized country-level Sharpe ratios also differ materially within the holdout: Israel delivers the highest single-country Sharpe (≈ 0.91) whereas Turkey is the weakest (≈ 0.24), reflecting differences in factor-signal persistence and local macro shocks rather than systematic strategy failure.

Country-weighted transaction costs. The 45-bp one-way transaction-cost assumption used throughout the paper is portfolio-weighted across EM countries using the Domowitz–Glen–Madhavan (2001) per-country estimates. The holdings-weighted average is approximately 44.6 bps; India—at roughly 18% of portfolio weight and among the lowest-cost markets in the sample—pulls the weighted average below the simple cross-country mean, while small frontier- adjacent markets contribute a higher marginal cost but only a small fraction of traded notional. The sensitivity analysis in Section 6 (Figure 6) brackets plausible deviations from this central estimate.

B Single-Factor Baseline

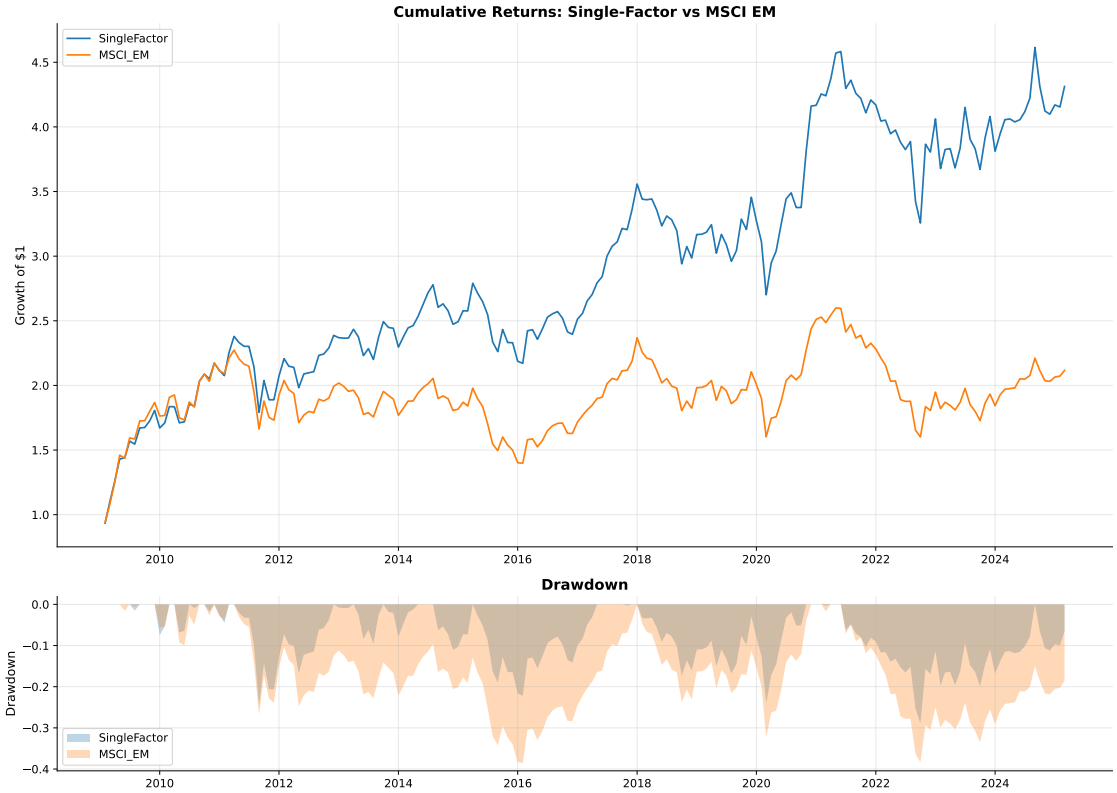

Figure 14: Cumulative returns of single-factor industry portfolios, 2009–2025. Cross-industry performance dispersion (range exceeds 300%) motivates cross-industry portfolio construction. Industries are selected by the factor with the highest rolling 60-month IC.

C Multi-Factor Composition Detail

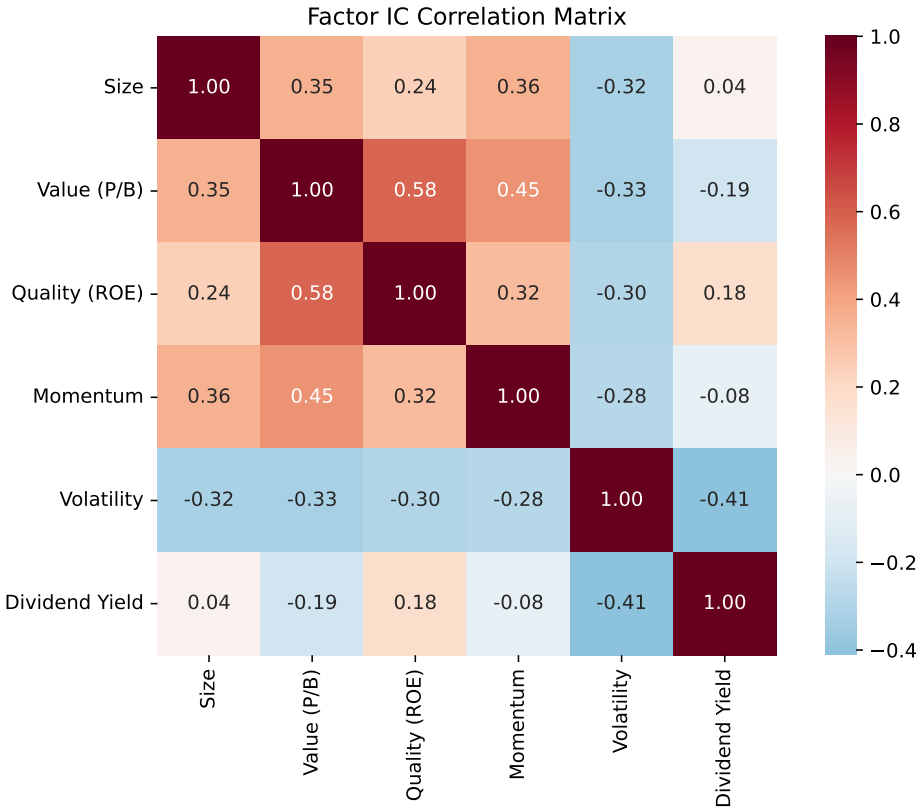

Figure 15: Pairwise IC correlations across six factors. Low inter-factor correlations, especially the negative momentum–value relationship (-0.31), support multi-factor combination. Diversified composites exhibit lower IC volatility than any individual factor.

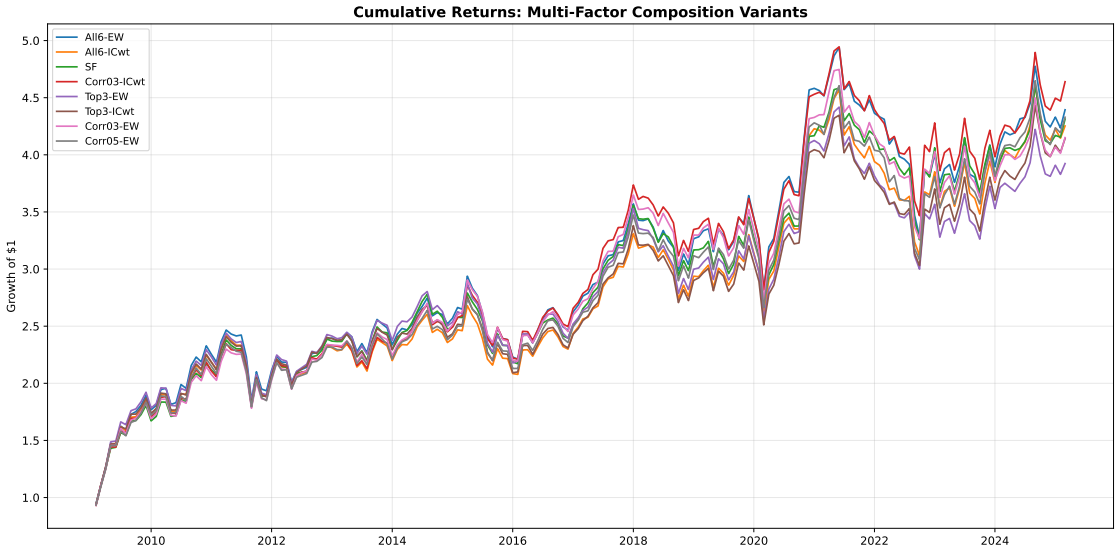

Figure 16: Cumulative returns of eight multi-factor composition variants, 2009–2025. All6-EW (thick line) performs competitively across the full period without any data-dependent tuning. IC-weighted and filtered composites show higher in-sample Sharpe but less stability.

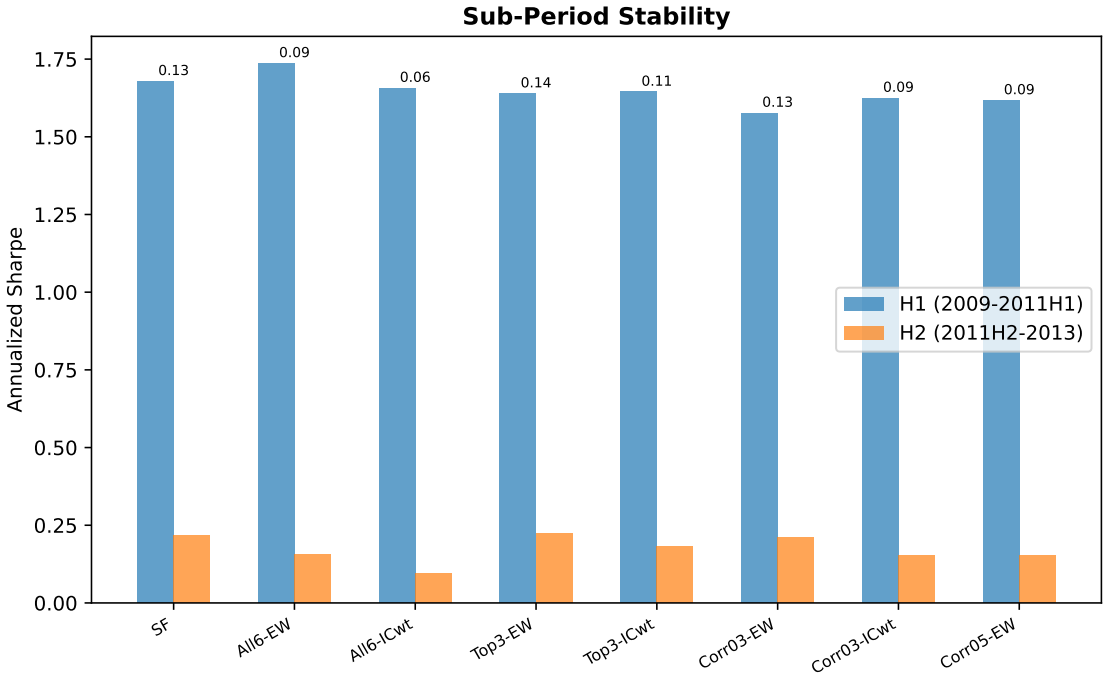

Figure 17: Multi-factor composition subperiod stability. All6-EW maintains competitive Sharpe across multiple sub-periods; more complex methods show higher variance.

D Portfolio Construction Detail

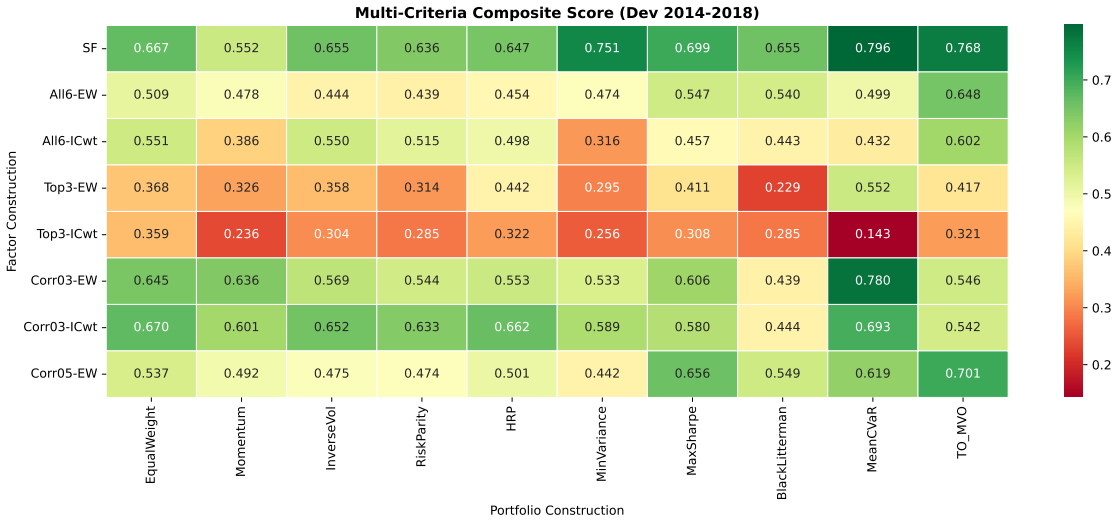

Figure 18: Multi-criteria composite scores for all 80 factor-PC combinations. TO_MVO consistently ranks highest regardless of the factor composite used. The composite score balances performance, stability, parsimony, TC robustness, and turnover.

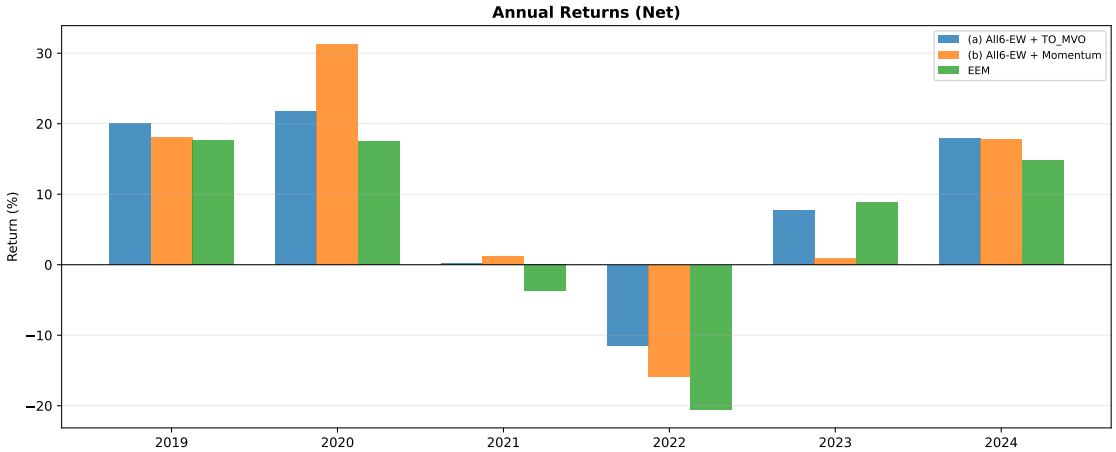

Figure 19: Annual holdout returns by strategy. TO_MVO outperforms EEM in five of seven calendar years. Underperformance in 2022–2023 coincides with the more difficult stock-selection environment discussed in Section 7.

Figure 20: TO_MVO parameter sensitivity. Sharpe ratio and turnover as functions of risk aversion λ and turnover penalty γ. Parameters are calibrated via time-series cross-validation on 2014–2018 data: $\lambda = 2.5$, $\gamma = 0.001$.

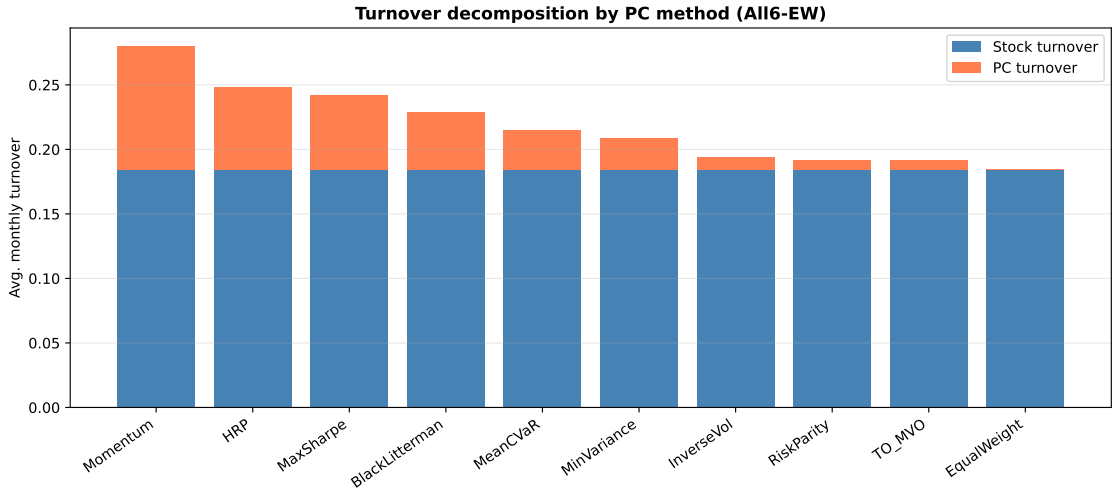

Figure 21: Turnover decomposition: stock-level vs. PC-level. Stock-level turnover (18.4%/month) is identical across all allocation methods. TO_MVO's L1 penalty reduces the PC-level component from 9.6% (Momentum) to 0.7%/month—a 93% reduction at the allocation layer.

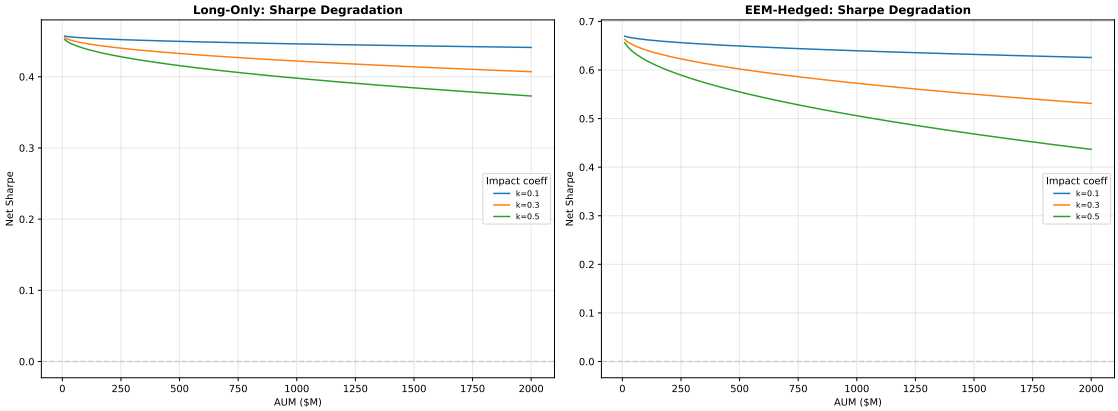

Figure 22: Sharpe ratio degradation with AUM. The hedged strategy maintains Sharpe > 0.700 up to approximately \$50M, declining to 0.641 at \$100M. Practical capacity (10th-percentile ADV at 10% participation) is approximately \$10M; median capacity is \$51M.

E Attribution and Hedge Instrument Detail

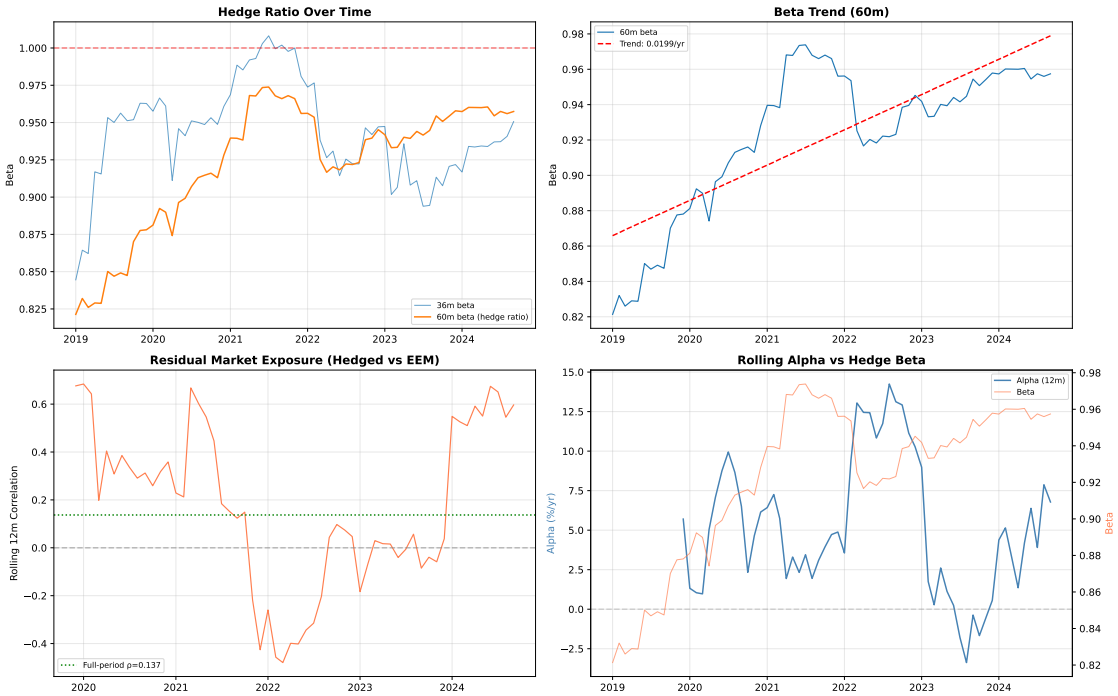

Figure 23: Rolling hedge ratio and residual market exposure, 2019–2024. The 60-month OLS beta fluctuates between 0.80 and 1.10. Average residual beta (post-hedge) is 0.049, indicating effective market neutralization within the matched sample.

Multi-window beta robustness. To verify that the hedge is not sensitive to a single choice of estimation window, we recompute the hedged series using 24-, 36-, 48-, 60-, and 72-month rolling betas, as well as an equal-weighted blend across these five windows. All six variants deliver

essentially identical holdout performance, with OOS Sharpe clustered in a tight $[0.78, 0.79]$ range; the equal-weighted blend attains Sharpe $= 0.792$, marginally above the baseline 60-month hedge (0.791). Residual beta is small and window- invariant. We therefore retain the simpler 60-month specification as our baseline and treat the multi-window blend as a robustness check rather than a standalone enhancement.

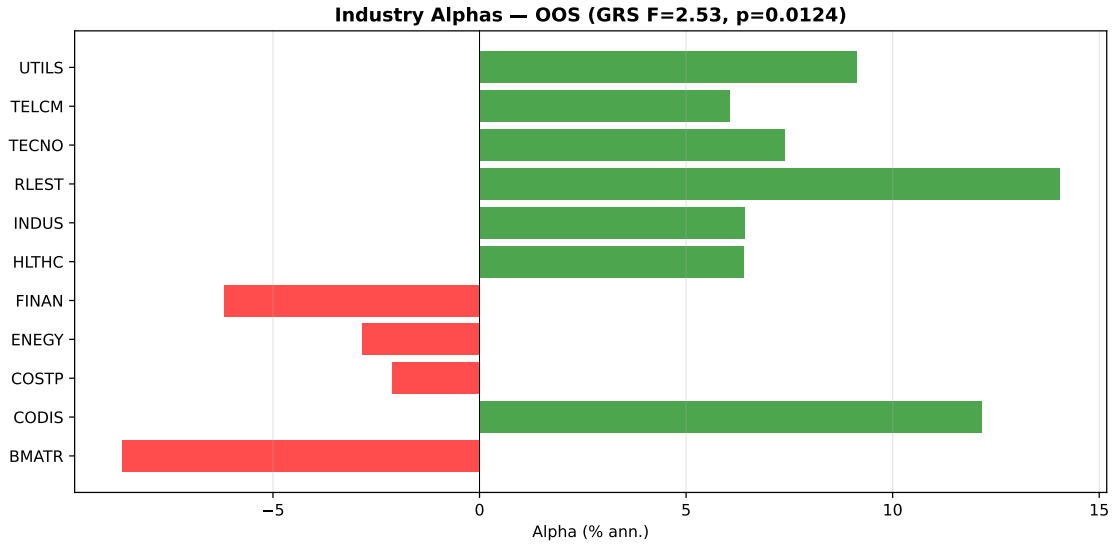

Figure 24: Industry-level OOS alphas (FF5+Mom regressions, $n = 69$). The GRS test ($F = 2.527$, $p = 0.012$) rejects jointly zero industry intercepts. Real Estate, Consumer Discretionary, and Utilities are the largest alpha contributors; Basic Materials and Financials detract.

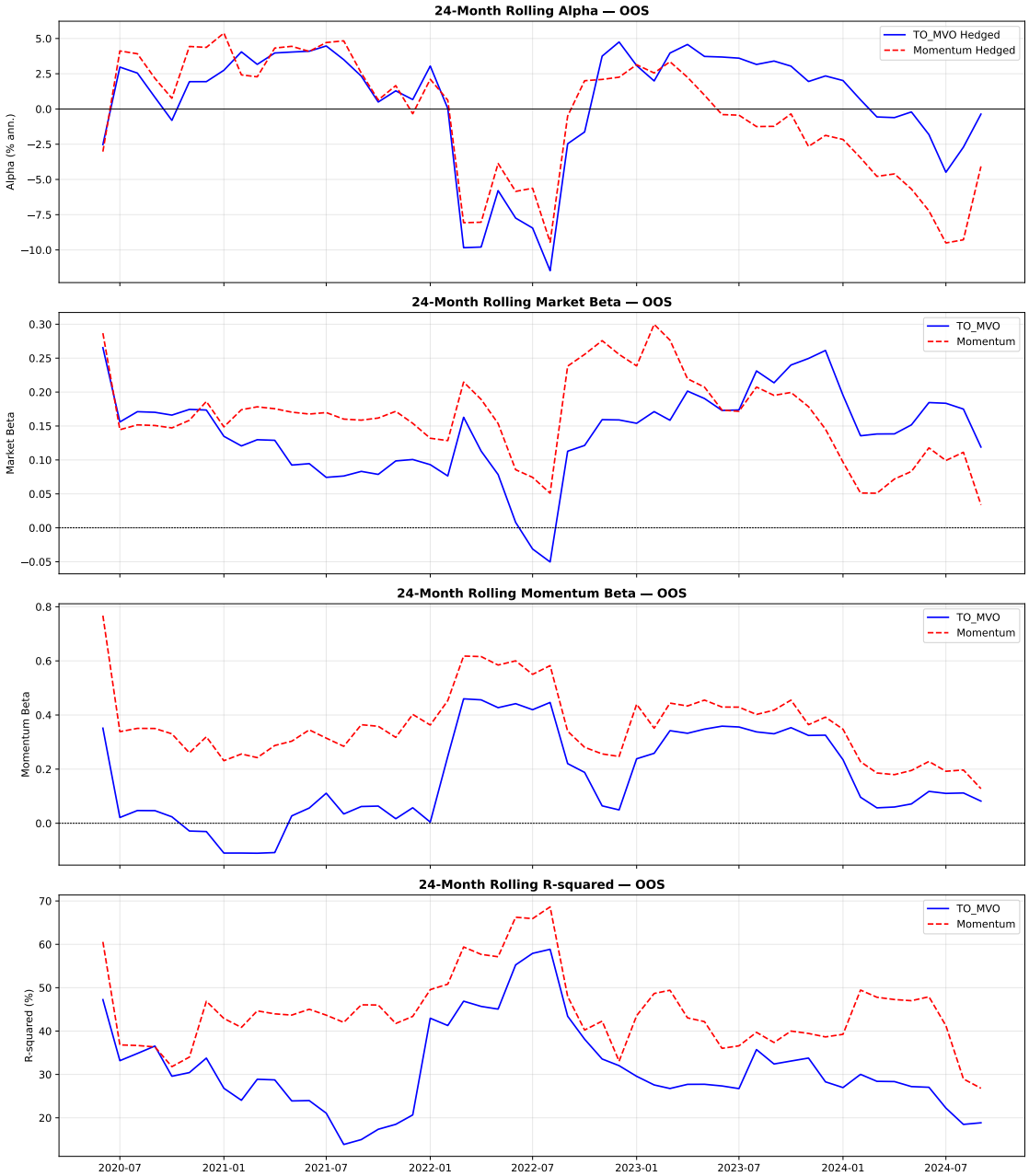

Figure 25: Rolling 36-month factor betas, TO_MVO long-only, OOS period. Factor loadings estimated over rolling 36-month windows. Market beta (top) is stable around 1.05–1.10. The negative HML and positive CMA loadings are persistent; RMW declines gradually from 2020 onward, consistent with the development-to-OOS shift in Table 4.

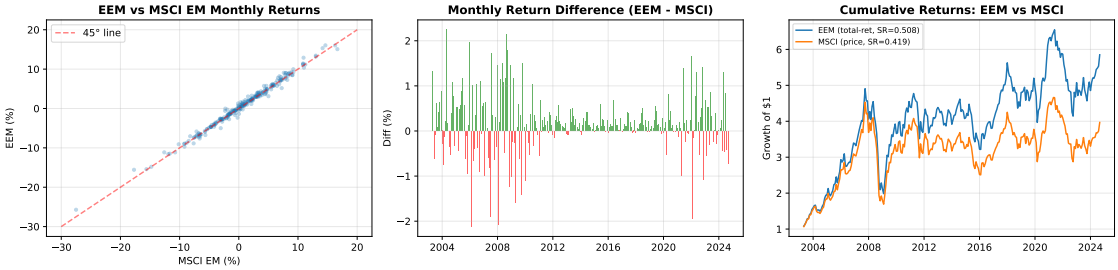

Figure 26: EEM total return vs. MSCI EM price index. The growing wedge ($\approx 1.8\%$/year) reflects cumulative dividends excluded from the price index. Using the price index as the hedge instrument would tend to overstate Jensen's alpha by this amount.

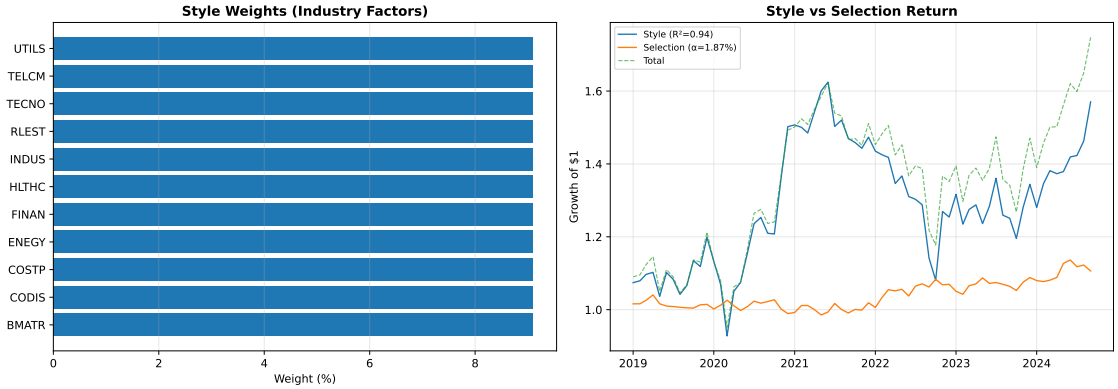

Figure 27: Sharpe style analysis: long-only TO_MVO. Style weights confirm the strategy's growth tilt (negative value loading) and moderate quality and size exposures, consistent with the factor model loadings in Table 3.

F FX Decomposition

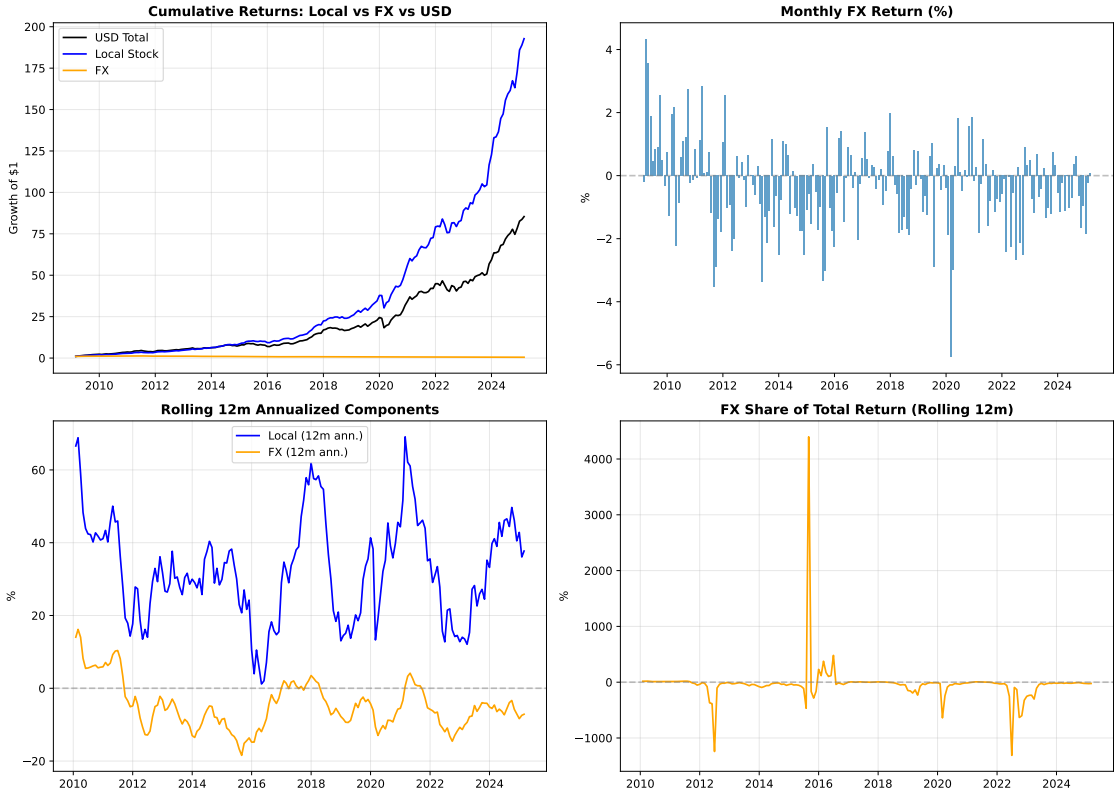

Figure 28: FX return decomposition. Local stock-selection returns contribute approximately 110% of USD total returns; EM currency depreciation subtracts approximately 1.7%/year. The beta hedge against EEM naturally neutralizes most FX risk since EEM holds the same currency basket.

Averaged across the OOS panel, the USD drag from EM currency weakness is modest but persistent at roughly 5–6 basis points per month. The drag is, however, highly regime-dependent: in 2023, for example, the long-only strategy returned approximately +40% in local terms but only +33% in USD, implying a $\sim$ 7-percentage-point FX headwind concentrated in that single year. Because EEM holds the same currency basket, the EEM beta hedge absorbs most of this drag at the portfolio level; the residual currency exposure that survives hedging is therefore small relative to the active stock-selection return.

G Holdings Diagnostics: Additional Figures

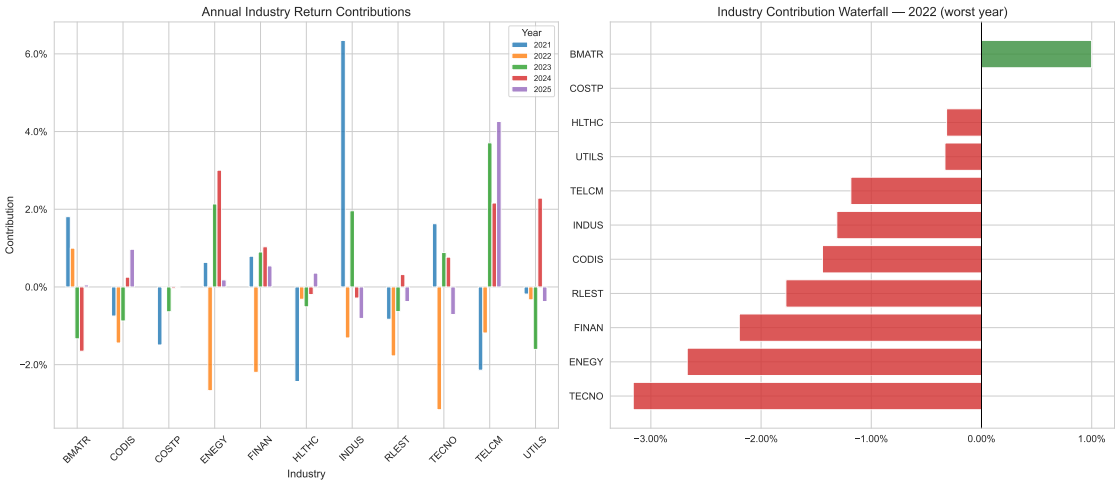

Figure 29: Industry contribution waterfall for the worst drawdown period, 2022H2–2023H1. Returns are decomposed by GICS industry. The loss is distributed across multiple industries, ruling out a single-sector event.

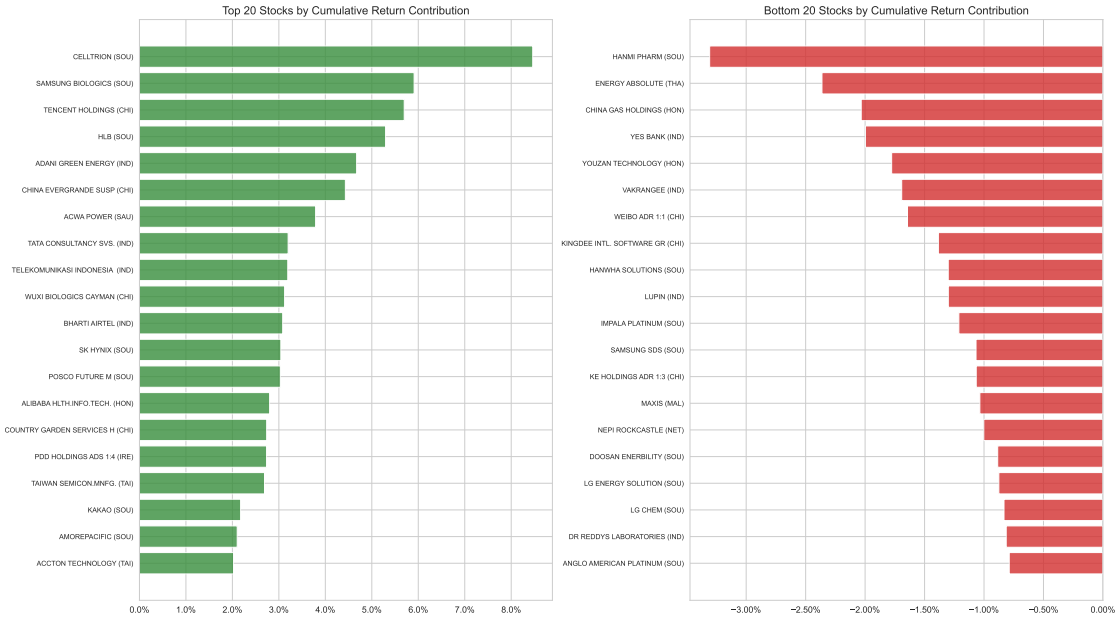

Figure 30: Top 20 and bottom 20 stocks by cumulative return contribution, 2021–2025. Top contributors are concentrated in Technology (Taiwan, Korea) and Consumer Discretionary (India). Bottom contributors reflect concentrated positions in lagging EM sectors during the 2022–2023 episode.

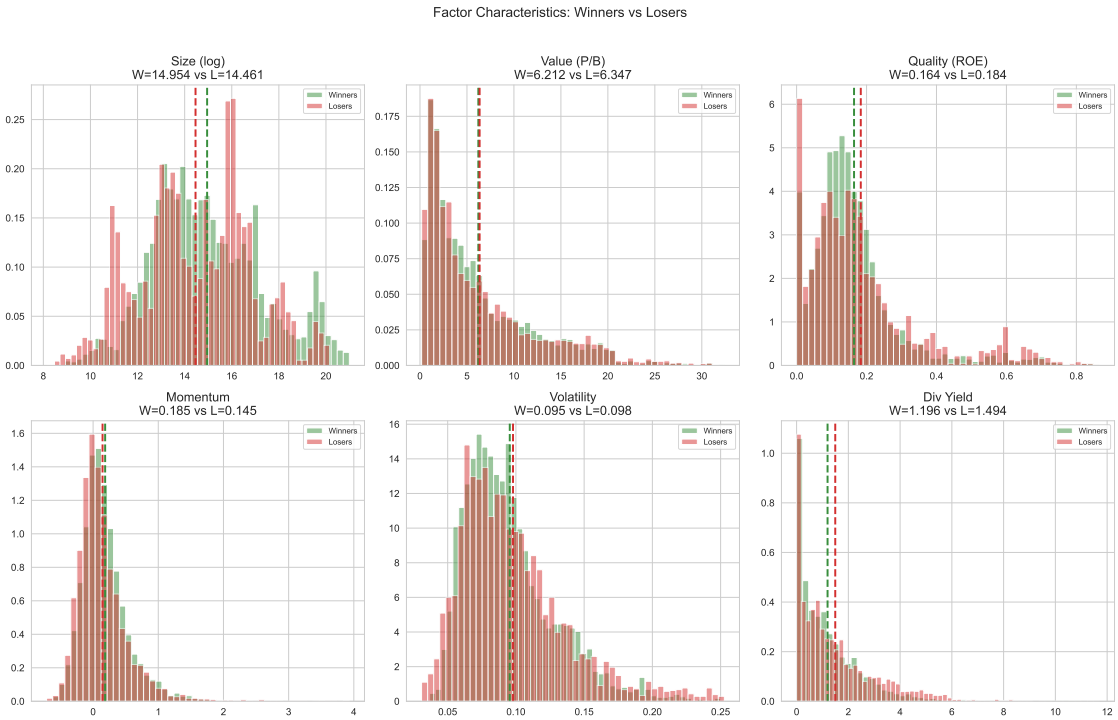

Figure 31: Factor characteristics of winning vs. losing stocks. Top-contributing stocks score higher on quality (ROE) and lower on volatility—consistent with the factor composite's intended signal. The characteristic gap widens in calmer periods and narrows during the 2022–2023 weakness.

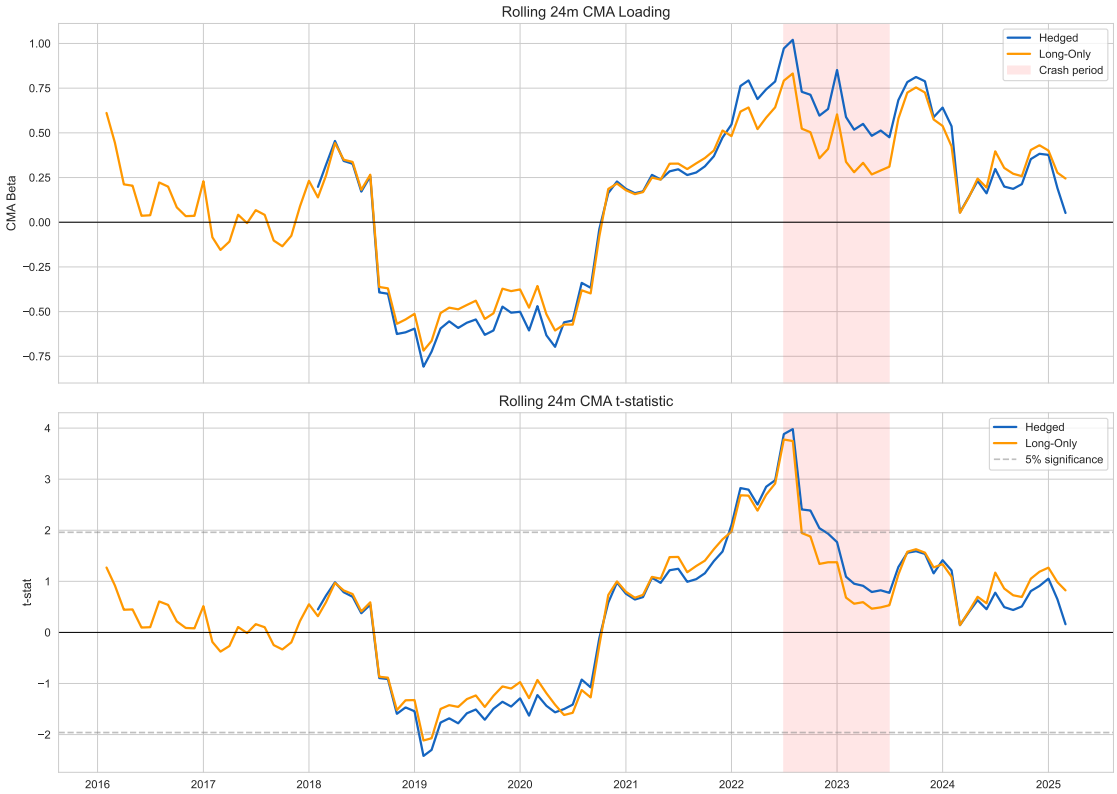

Figure 32: Rolling 24-month multivariate CMA loading by strategy, 2019–2024. The FF5+Mom CMA coefficient remains positive across much of the sample for both long-only and hedged returns, but its magnitude varies over time. This time variation reinforces the need to interpret the coefficient cautiously in light of CMA–HML collinearity.

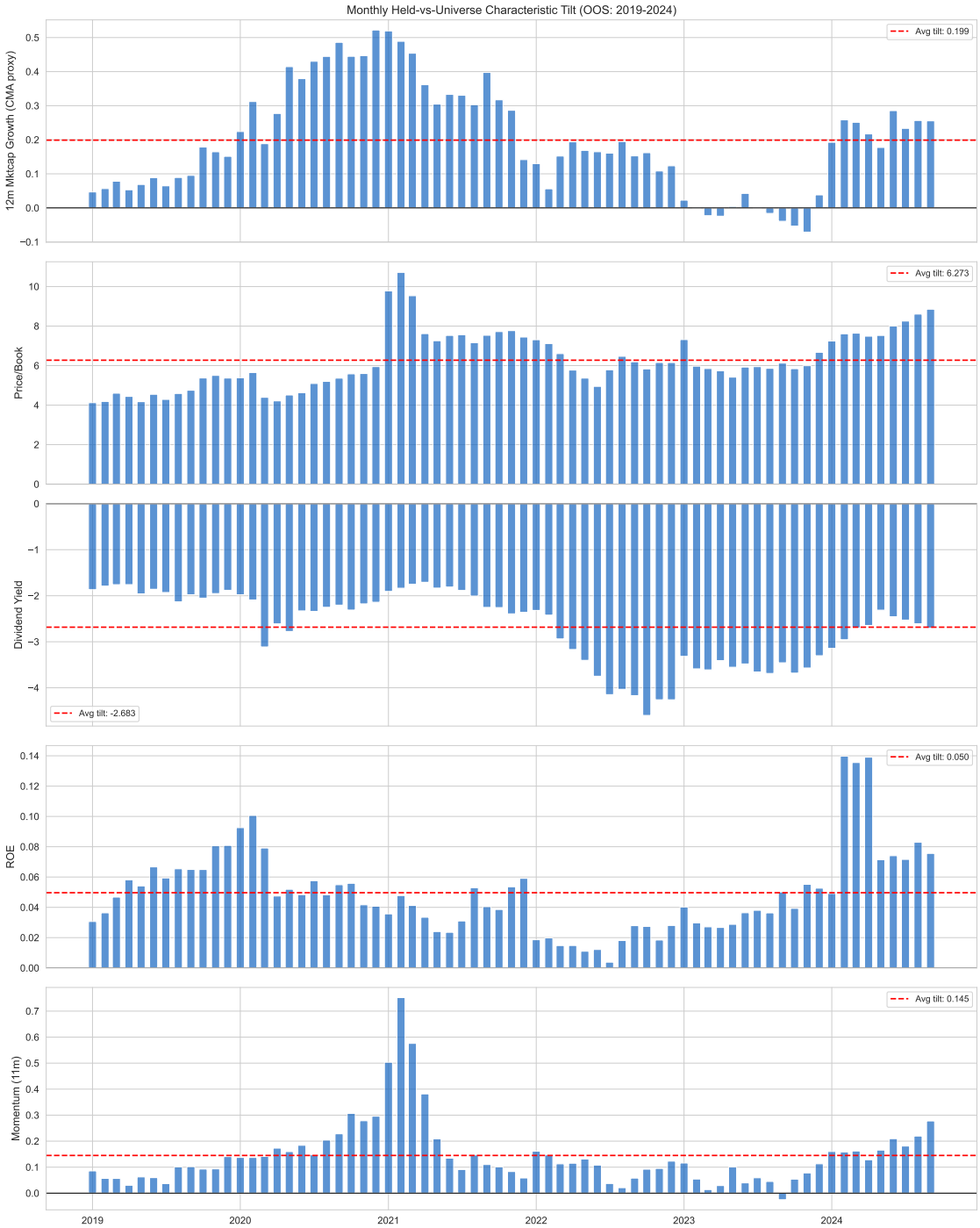

Figure 33: Portfolio factor tilts over time. Monthly average factor scores of held stocks relative to the full universe. Within the available 2021–2024 holdings panel, the strategy maintains broadly similar tilts toward size, value (growth), quality, and momentum before and after the 2022–2023 drawdown.

H US Diversification: Supplementary Results

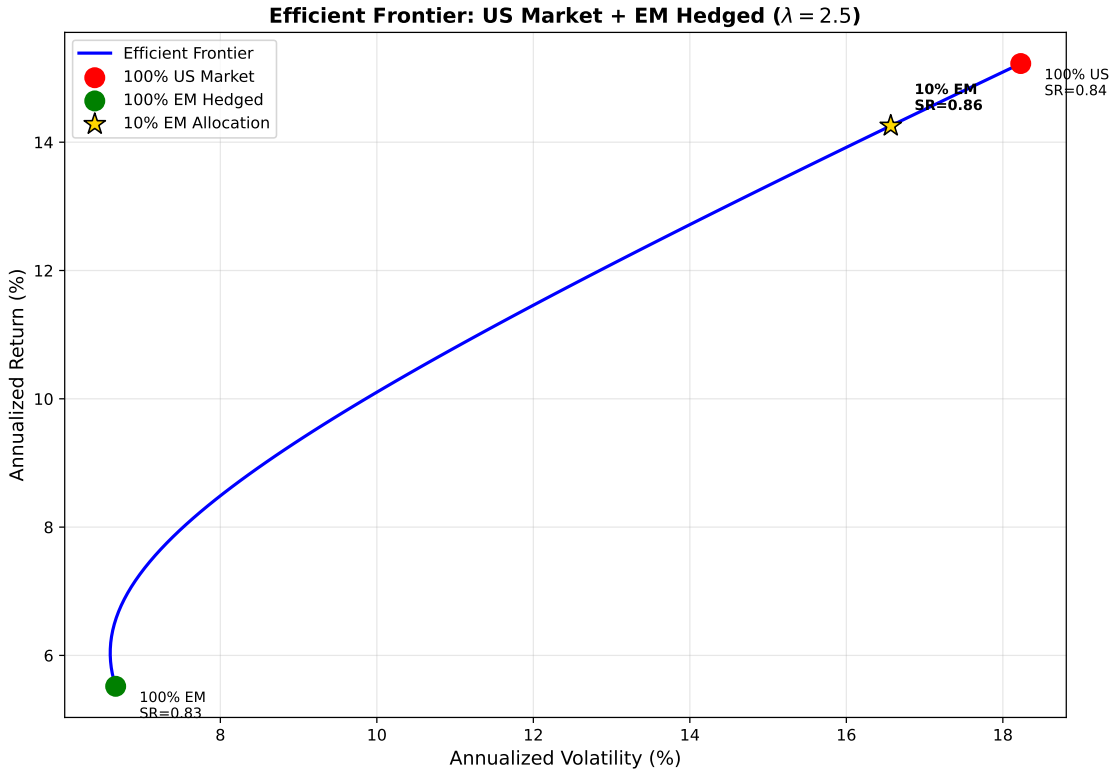

Figure 34: Efficient frontier: US equities + EM hedged strategy. A 10% EM allocation reduces portfolio volatility by 9.1% and improves the Sharpe ratio from 0.835 (100% US) to 0.860. The leftward curvature at small EM weights reflects the low 0.223 correlation.

Figure 35: Rolling 24-month alpha vs. US factors. Alpha relative to US Fama-French five factors plus momentum is positive in many rolling windows, suggesting that the diversification benefit is not concentrated in a single episode.

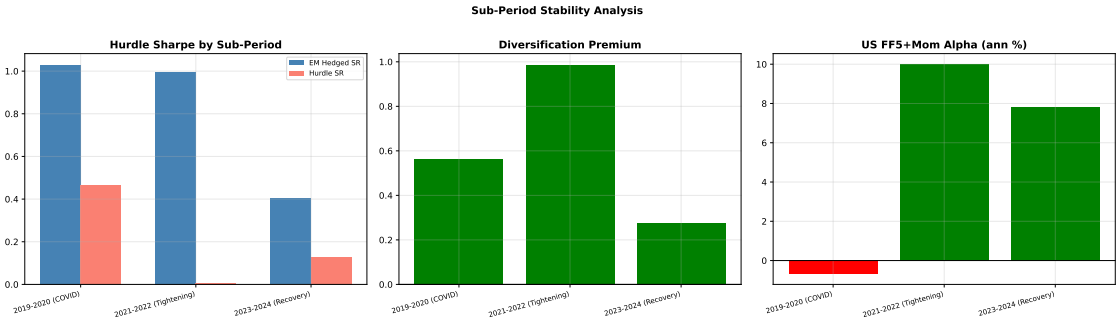

Figure 36: Hurdle Sharpe performance by sub-period. The strategy passes the hurdle Sharpe test in every sub-period (2019–2020, 2021–2022, and 2023–2024 through September), suggesting that the diversification benefit is not driven by a single favorable window.